

Specific Aims

Alzheimer's Disease (AD) impairs not only cognitive processes, such as memory, but also perceptual processes, such as vision.^{1,2} Memory and visual deficits in AD are strongly correlated but the neural mechanisms of this link are unclear.³⁻⁵ The proposed studies investigate memory for visual stimuli, which declines substantially in early AD, predicting dementia more than a decade before diagnosis.⁶ We propose 3 functional MRI (fMRI) studies, each including 3 groups of participants: (1) individual with *amnestic Mild Cognitive Impairment (aMCI)* (including both *aMCI-single domain* and *aMCI-multi domain*^{7,8}), who have a high probability of progressing to AD; (2) healthy older adults (**hOAs**), and (3) healthy younger adults (**YAs**). The studies separately measure the effects of aMCI, by comparing aMCI individuals to hOAs, and the effects of healthy aging, by comparing hOAs to YAs. Whereas most prior memory fMRI studies with aMCI individuals have focused on *memory processes*, we propose to examine, for the first time, the effects of aMCI on *memory representations*.

We focus on memory representations for visual, nameable objects (chairs, cars, etc.). These stimuli are processed along the ventral visual pathway, with more posterior *occipito-temporal cortex (OTC)* regions analyzing primarily simpler visual features (e.g., edges, corners) and more anterior OTC regions, analyzing more complex, conjunctive visuo-semantic features (integrated objects, object categories). Memory representations for these visual and semantic features are stored within the same OTC regions that processed them: *visual feature representations* are stored primarily in posterior OTC, and *semantic feature representations*, mainly in anterior OTC. Memory representations are also stored in the *medial temporal lobe (MTL)*, which also has a posterior-visual to anterior-semantic organization. Compared to cortical representations, MTL representations are more relational, binding item-item and item-context associations. However, the current project emphasizes visual vs. semantic representational differences that cut across OTC and MTL, and hence, we use the expressions “posterior OTC/MTL” and “anterior OTC/MTL.” Each of the three proposed studies tackles **2 specific aims**.

Aim 1. Investigate the effects of healthy aging and aMCI on visual vs. semantic features of memory representations. fMRI studies have shown that visual feature representations in posterior OTC/MTL become less distinct (*dedifferentiated*) in healthy aging.⁹⁻¹⁴ Yet, despite this decline, hOAs have no difficulty with basic object identification. Using novel *representational similarity analyses (RSA)* and *encoding-retrieval similarity (ERS)* analyses, we have found that semantic feature representations in anterior OTC/MTL are spared in hOAs. This finding is consistent with evidence that semantic processing and anterior OTC/MTL function is relatively preserved in hOAs,¹⁵⁻²² who may rely on spared semantics to compensate for perception and memory deficits.²³⁻²⁵ In contrast, aMCI individuals are impaired in semantic processing²⁶⁻²⁸ and anterior OTC/MTL function.²⁹⁻³³ Thus, semantic feature representations in anterior OTC/MTL are likely to be more impaired in aMCI individuals than hOAs but direct evidence is not available. In all 3 proposed studies, we test the hypothesis that visual feature representations in posterior OTC/MTL are impaired in both aMCI individuals and hOAs, whereas semantic feature representations in anterior OTC/MTL are impaired in aMCI individuals but not in hOAs (Hypothesis 1). Although visual representations are probably more impaired than semantic representations in aMCI individuals, semantic representations are more informative because they are likely to be impaired in aMCI individuals but they are largely spared in hOAs. This hypothesis predicts a dissociation between aMCI and healthy aging, and hence it has implications for understanding and diagnosing early AD. While testing Hypothesis 1, Studies 1-3 studies investigate 3 different factors contributing to visual feature representation deficits during encoding: **Study 1** examines the role of a **visual input deficit** by manipulating stimulus degradation; **Study 2** investigates the role of an **associative memory deficits** by varying binding demands; and **Study 3** examines the role a **top-down attention** deficits by manipulating attentional cueing.

Aim 2. Examine the effects of semantic support on memory representations in hOAs and aMCI. The *semantic support effect* refers to the reliable finding that conditions that allow and/or promote the use of preexistent semantic knowledge tend to attenuate cognitive deficits in older adults.^{23,34-36} The semantic support effect is weaker in aMCI individuals,³⁷⁻³⁹ consistent with their semantic processing deficits²⁶⁻²⁸ and anterior OTC/MTL dysfunction.²⁹⁻³³ We hypothesize that the semantic support will enhance the quality of memory representations and subsequent memory to a greater extent in hOAs than in aMCI individuals (Hypothesis 2). Such a finding has implications for distinguishing the neural mechanisms of healthy and pathological aging and for the development of memory training methods for hOAs and aMCI individuals. Studies 1-3 investigate 3 different forms of the semantic support effect during encoding: **Study 1** examines semantic processing of individual stimuli during a **deep encoding** task; **Study 2** investigates the encoding of new associations as a function of **schema consistency**, and **Study 3** examines the focusing of encoding processes by **valid conceptual expectations**.

In sum, the proposed studies systematically investigate the effects of healthy aging and aMCI on memory representations for visual stimuli. They use novel RSA and ERS measures and examine fundamental questions about the neural mechanisms of memory. They are also significant, including potential implications for the diagnosis of early AD and the development of memory training methods.

A. Significance.

Our **overarching goal** is to investigate the effects of healthy aging and *amnesic mild cognitive impairment* (**aMCI**) on the neural mechanisms of memory for visual objects. aMCI is a transitional stage between healthy aging and AD.⁴⁰ (Note: here and elsewhere, we use “aMCI” instead of “amCI” for points or references that are not specific to aMCI and may apply to aMCI in general). Individuals with aMCI exhibit an impairment in memory and other cognitive abilities, with minimal decline in activities of daily living, and no dementia.^{7,8} Although aMCI does not guarantee development of dementia, most aMCI individuals progress to clinical AD (~80% over 6 years^{7,40}). Because most aMCI individuals have preclinical AD, aMCI individuals represent an ideal focus to investigate the neural mechanisms of early AD.

Investigating memory in the visual modality is significant because both *healthy older adults* (**hOAs**) and aMCI individuals show substantial visual decline. Healthy aging is associated with significant visuo-perceptual deficits (e.g., in contrast sensitivity, color vision, and shape perception^{41,42}), consistent with age-related decline of early^{43,44} and late visual cortex.⁴⁵ Visual deficits in hOAs are strongly correlated with cognitive decline,^{3,4} and can predict this decline several years in advance.⁵ Individuals with aMCI also display substantial visual deficits^{1,2,46,47} (e.g., in acuity,^{48,49} contrast sensitivity,⁵⁰ color⁵¹, motion perception),^{52,53} which are strongly associated with their cognitive decline. In fact, up to 50% of the variance in AD-related cognitive decline can be explained by visual deficits.⁵⁴ Visual deficits in AD and aMCI are consistent with the accumulation of plaques and tangles in visual cortex,^{55,56} which can occur in this region even before than in the *medial temporal lobes* (**MTL**).⁵⁷ Memory for visual stimuli, which depends on both visual and memory functions, is very sensitive to the effects of early AD, and can predict AD more than a decade before diagnosis,⁶ as well as conversion from aMCI to AD with high specificity.⁵⁸ Thus, our overarching goal has implications for understanding and diagnosing early AD.

Our **two specific aims** are also significant. Our **Aim 1** is to *investigate the effects of healthy aging and aMCI on visual vs. semantic features of memory representations*. Visual feature representations in the posterior *occipito-temporal cortex* (**OTC**) are impaired by aging, as illustrated by the finding that posterior OTC activations for different visual stimuli (faces, scenes, etc.) become less distinct in old age (*age-related dedifferentiation*).⁹⁻¹⁴ However, our preliminary data show that semantic feature representations in anterior OTC/MTL are relatively preserved in hOAs, consistent with the sparing of semantic knowledge in old age. In contrast, aMCI individuals show impairments in semantic memory⁵⁹⁻⁶³ and anterior OTC/MTL function.²⁹⁻³³ Thus, we hypothesize that *visual feature representations in posterior OTC/MTL are impaired in both aMCI and hOAs, whereas semantic feature representations in anterior OTC/MTL are impaired in aMCI individuals but not in hOAs* (**Hypothesis 1**). Such dissociation would be highly novel and expedient clinical tool for distinguishing between healthy aging and aMCI, and hence, it has implications for early AD diagnosis.

Our **Aim 2** is to *examine the effects of semantic support on memory representations in hOAs and aMCI*. The *semantic support effect* refers to the reliable finding that conditions that promote the use of preexistent semantic knowledge tend to attenuate cognitive deficits in older adults.^{23,34-36} The semantic support effect is weaker in aMCI individuals,³⁷⁻³⁹ in line with their semantic memory deficits.²⁶⁻²⁸ We hypothesize that *semantic support will enhance the quality of memory representations and subsequent memory to a greater extent in hOAs than aMCI individuals* (**Hypothesis 2**). Investigating the neural mechanisms of the semantic support effect is significant for at least two reasons. **First**, because the semantic support effect is reduced by aMCI, differences between hOAs and aMCI individuals become more pronounced under such supportive conditions.³⁸ Thus, the semantic support effect exposes the limits of aMCI individuals’ abilities, facilitating the distinction between healthy and pathological aging. **Second**, the semantic support effect is often used in memory training programs in hOAs and aMCI individuals.^{64,65} Thus, understanding the neural mechanisms of the semantic support effect and why it is reduced in aMCI individuals could help in the development of new memory training methods. These new methods will not prevent AD but they will provide aMCI patients with useful strategies for preserved memory functioning. **C1.5**

B. Innovation

The proposed studies are innovative in several ways. **First**, they represent the first systematic investigation on the effects of healthy aging and aMCI on memory representations. Most fMRI studies on memory in aging and AD have focused on memory processes and have not examined the quality of the underlying representations used and generated by these processes. Yet, the quality of memory representations is at least as important for successful memory performance as the efficacy of memory processes. **Second**, the proposed studies use cutting-edge *representational similarity analyses* (**RSA**) and *encoding-retrieval similarity* (**ERS**) analyses that have only rarely been used to study the effects of healthy aging on memory, and never—to our knowledge—to study the effects of aMCI. Our RSA approach employs sophisticated modeling methods that can distinguish between visual and semantic features of memory representations, and can link them to performance in different subsequent memory tasks. Disentangling visual and semantic aspects of memory representations is critical, as healthy aging tend to impair visual processing^{41,42} but not semantics,⁶⁶⁻⁶⁹ whereas aMCI affects both.^{1,2,46,47,59-63} **Finally**, the proposed studies will be among the first to examine the interaction between *control* processes and memory

representations. While many univariate fMRI studies of memory have investigated control processes (encoding elaboration, retrieval monitoring, etc.) and several recent studies have assessed memory representations, there is a dearth of studies examining the interaction between control processes and representations, particularly in regard to aging and aMCI. In sum, the proposed studies are novel regarding their topic, their representational methods, and examination of control-representation interactions.

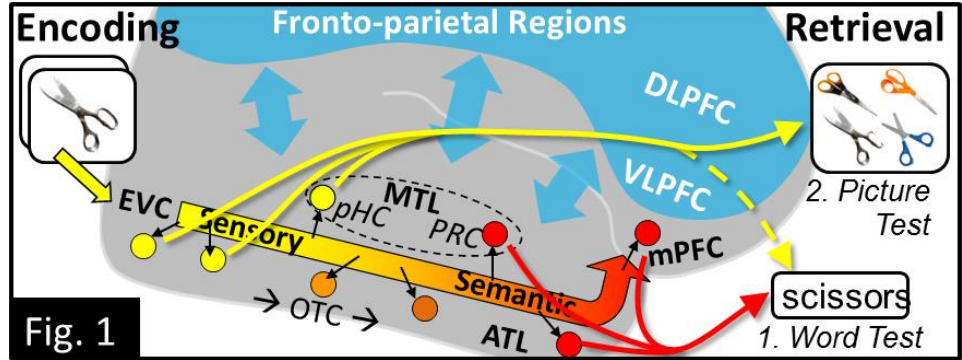
C. Approach

Our Approach is described across 6 sections: (1) General background, (2) Aim 1 Background, (3) Aim 2 Background, (4) Methods shared across studies, (5) Studies 1-3 and exploratory combined analyses, and (6) Detailed Methods. Preliminary studies are included in multiple sections and marked with a box.

1. General background

A Model of object memory.

The model in **Fig. 1** only describes the components investigated in the current project (i.e., other regions and processes are involved). Many assumptions of the model are topics of debate, but for the sake of space, we describe them as a matter of fact (e.g., instead of saying “some evidence suggests that Region X mediates Y,” we just say “Region X mediates Y”). From early visual cortex, visual object features are gradually processed (yellow-orange gradient) along the occipito-temporal cortex (OTC), with more **posterior** OTC regions analyzing relatively simpler visual features (e.g., edges, shapes), and more **anterior** OTC regions, processing relatively more complex, conjunctive visuo-semantic features (e.g., integrated objects, object categories). For simplicity, we call the former “visual features” and the latter, “semantic features,” but the difference is graded, not sharp. Semantic processing reaches its confluence at the anterior temporal lobe (ATL), where visual and semantic features are integrated, and which feeds into the medial prefrontal cortex (mPFC), where **knowledge schemas** are processed and stored,^{55,56} and left ventrolateral prefrontal cortex (VLPFC), which mediates semantic control processes.⁷⁰



Memory representations (yellow-red circles) are stored within the same regions performing each kind of analysis: visual feature representations, in posterior OTC; semantic feature representations, in anterior OTC. Memory representations are also stored in MTL, which largely abides this posterior-anterior organization.^{71,72} The representations stored in **posterior** MTL regions, such as the posterior hippocampus, are more visual whereas those stored in anterior MTL, such as the anterior hippocampus and *perirhinal cortex* (PRC), are more semantic.⁷² The posterior hippocampus has been linked to fine visual discriminations, and the anterior hippocampus, to more abstract discriminations.⁷² Resting-state functional connectivity studies show that the posterior hippocampus is connected to posterior cortical regions, and the anterior hippocampus and PRC to anterior brain regions.^{71,73,74} The difference between cortical and MTL memory representations is complex but for the present purposes we assume that MTL memory representations are more relational (binding item-item and item-context representations). However, because we emphasize posterior (visual) vs. anterior (semantic) differences that exist both in OTC and MTL, we often write “posterior OTC/MTL” and “anterior OTC/MTL.”

OTC/MTL processes and representations interact with processes in fronto-parietal regions, including VLPFC and dorsolateral PFC (DLPFC). Although the emphasis of this project is on OTC/MTL representations, we also investigate how these representations relate to encoding control processes in VLPFC and DLPFC. For the sake of space, we do not cover data and predictions regarding parietal regions in this project.

The model assumes that for a single visual object, multiple memory representations are stored: posterior OTC/MTL stores visual feature representations while anterior OTC/MTL stores semantic feature representations of the same object. During retrieval, some of these multiple memory traces are *reactivated*, allowing memory for different aspects of original object. This assumption is supported by abundant evidence from behavioral memory studies.⁷⁵⁻⁷⁹ In Bahrack and Boucher’s^{78,79} behavioral studies, for example, participants encoded object pictures (e.g., a picture of a scissor), and memory for each object was tested twice (see right side of **Fig. 1**), first in word recognition test (e.g., “scissors”), and then in a picture forced-choice test (e.g., several scissor pictures). Remarkably, performance in the two tests was uncorrelated, which provides strong evidence that they are mediated by different memory representations. The authors’ explanation is that the word test was mediated primarily by semantic feature representations, and the picture test, mainly by visual feature representations. In our model, these effects would correspond to the solid red and yellow arrows in **Fig. 1**. However, if participants remember the image of an object in the word test, then this test becomes sensitive not only to semantic but also to visual features (dashed yellow arrow). In the proposed studies, we use this kind of “hybrid” word task, which allows us to assess the retrieval of both types of representations in the same task.

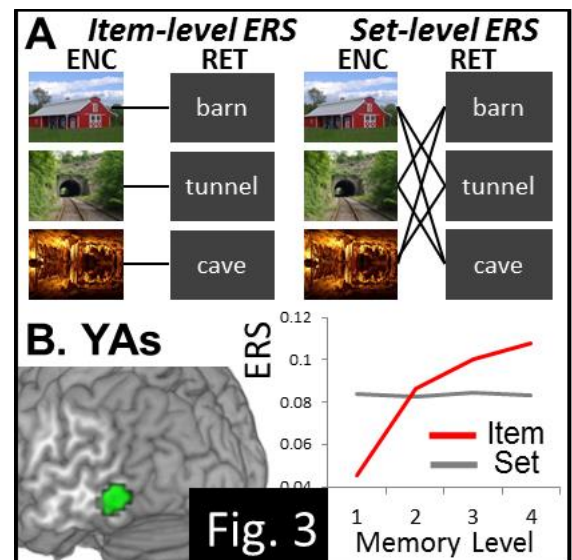
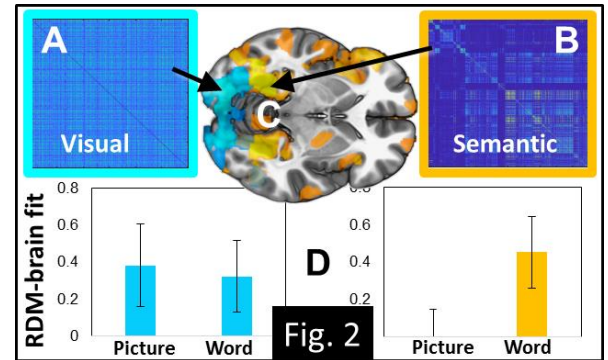
Model-based Representational Similarity Analysis (RSA). First, a **representational dissimilarity matrix (RDM)** is created. Each cell of the RDM contains the similarity between every pair of stimuli (e.g., two object pictures) in one particular dimension. Complex stimuli such as everyday objects have multiple dimensions. For example, in a *visual dimension*, a basketball is similar to a pumpkin but not to a golf club, whereas in a *semantic dimension*, the basketball is similar to the golf club but not to the pumpkin. Thus, for the same set of objects one can have multiple RDMs: a visual RDM, a semantic RDM, etc. Second, for each *region-of-interest (ROI)*, a **brain similarity matrix** is computed in which each cell indicates the similarity in activity patterns elicited by each pair of stimuli. Finally, RDM and brain similarity matrices are correlated with each other, yielding an **RDM-brain fit**, which indicates the sensitivity of each ROI to each stimuli dimension. This RSA approach has been used in semantic memory fMRI studies to investigate the sensitivity of brain regions to visual and semantic features.⁸⁰⁻⁸⁴ Going beyond these studies, we examine how this sensitivity *predicts* subsequent episodic memory.

For example, in a preliminary RSA study,⁸⁵ 22 YAs encoded 300 objects belonging to 12 different categories. A **visual RDM (Fig. 2-A)** used visual similarity values from a *deep convolutional neural network (DNN)* model of visual cortex, and a **semantic RDM (Fig. 2-B)** used similarity values from semantic association norms (e.g., a *dog* lives, eats, has legs; a *canary* lives, eats, flies, etc.). As illustrated by **Fig. 2-C**, the RDM-brain fit was significant ($p < 0.05$, FWE corrected) for the visual dimension in posterior OTC regions (blue) and for the semantic dimension in anterior OTC and PFC regions (orange). Critically, the RDM-brain fits for visual and semantic dimensions differentially predicted subsequent memory in word vs. picture tests. As shown by **Fig. 2-D**, the RDM-brain fit for the semantic dimension in anterior OTC/PFC (orange regions) predicted subsequent memory (remembered > forgotten) in the word test but not in the picture test, while the RDM-brain fit for the visual dimension in posterior OTC (blue regions) predicted memory in both tests. The contribution of visual representations to the word test can be interpreted as reflecting the use of visual imagery in this test. The sensitivity of the word test to both semantic and visual representations is an advantage for our goals because it allows us to examine the reactivation of both types of representations in the same test, which we measure using encoding-retrieval similarity (ERS).

Encoding-Retrieval Similarity (ERS). Memory representations are assumed to be stored in the same regions that process the stimuli and are activated during encoding,^{86,87} which are then re-activated during retrieval when the representations are accessed.⁸⁸ Whereas traditional reactivation fMRI studies measured the reactivation of sets of items,⁸⁹ we use ERS to measure the reactivation of individual items. For example, in a preliminary scene ERS study with YAs (Wing et al., 2015),⁹⁰ participants encoded scenes paired with labels (e.g., “barn”), and during retrieval, only the labels were presented (word test) and participants recalled the corresponding scenes, rating their memories for vividness. As shown by **Fig. 3-A**, we compared ERS for matching encoding-retrieval pairs (**item-level ERS**, e.g., barn-barn) to ERS for mismatching pairs (**set-level ERS**, tunnel-barn, cave-barn, etc.), which provide a baseline for ERS among different pictures. In posterior OTC, item-ERS (but not set-ERS) increased with memory success (**Fig. 3-B**), suggesting that this region stores scene-specific visual features that are reactivated during retrieval, mediating memory performance. As noted above, even if this test uses words as cues, because participants retrieve and rate visual images, the task is sensitive to both visual and semantic feature representations.

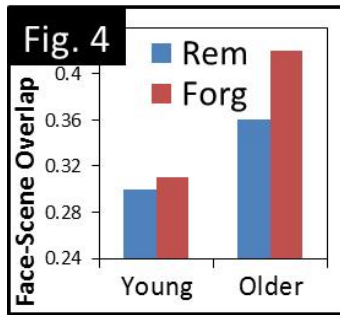
2. Aim 1: Effects of aging and aMCI on visual vs. semantic features of memory representations

Age-related dedifferentiation. fMRI studies have shown that in posterior OTC, activation patterns for different visual stimuli are less distinct in hOAs than YAs. This effect, known as *age-related dedifferentiation*,⁹ and has been replicated for different stimuli,⁹⁻¹⁴ and linked to deficits in several cognitive functions, including imagery,¹² working memory,⁹¹ and fluid intelligence.¹¹ It is unclear, however, if dedifferentiation of perceptual representations affects the quality of the memory representations. We found evidence for this effect in a reanalysis of an encoding fMRI study.⁹² The results showed that the overlap between face-sensitive and scene-sensitive OTC voxels (formula F from⁹³) was greater for hOAs than for YAs (**Fig. 4**, $p < 0.05$), indicating dedif-



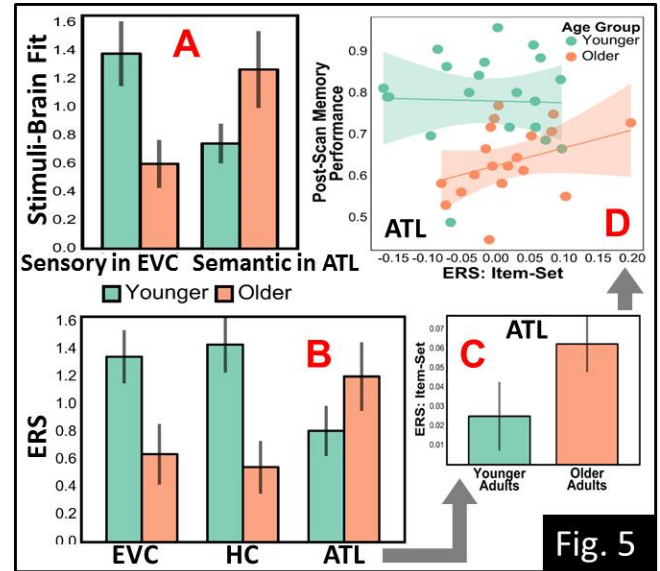
ferentiation. Critically, we found that in hOAs, the overlap was greater for subsequently forgotten than remembered items, suggesting that age-related dedifferentiation impairs memory representations.

Memory representations in hOAs. A few fMRI studies have used representational analyses to examine



memory representations at the item-level; and (3) we directly link RSA and ERS to memory performance; and (4) we compare hOAs to aMCI individuals.

To illustrate our approach, in a recent aging RSA/ERS study,⁹⁷ YAs and hOAs viewed scenes paired with labels, and during retrieval, they read the labels (word test) rating the memory of the scene. Again we relied on DNN to quantify the visual properties of these scenes,^{98,99}; early layers provided the visual RDM and late layers, the semantic RDM.^{100,101} The study yielded 3 main findings. **First**, as shown by **Fig. 5-A**, there was a cross-over dissociation between posterior and anterior OTC (age x ROI x representation type interaction: $p < .01$): the visual RDM-brain fit in posterior OTC (early visual cortex—EVC) was reduced in hOAs, consistent with age-related dedifferentiation, whereas the semantic RDM-brain fit in anterior OTC (ATL) was enhanced by aging. This is the first time this “hyperdifferentiation” effect is reported. These results suggest that visual representations in posterior OTC are impaired by aging, whereas semantic representations in anterior OTC are spared. Given that visual processing in posterior OTC feeds into semantic processing in anterior OTC, one may wonder how the latter can be spared if the former is impaired. A possible explanation is that semantic processing does not require complete, well-defined visual details, and can be achieved even if visual processing is mildly impaired, as in the case of hOAs. **Second**, consistent with the RSA finding, ERS in early visual cortex and posterior hippocampus was reduced by aging, whereas ERS in ATL was enhanced by aging (**Fig. 5-B**). The latter effect was also significant when the analysis was limited item-specific ERS (**Fig. 5-C**). **Finally**, the enhanced ERS effect in hOAs was beneficial for memory performance in this group, as indicated by a reliable correlation between ERS and post-scan memory performance (**Fig. 5-D**). This finding suggests that hyper-differentiation and enhanced ERS in ATL may serve a compensatory role. In sum, the study showed that, in contrast with the detrimental effects of aging on visual feature representations in posterior OTC/MTL, semantic features in anterior OTC were enhanced by aging, possibly reflecting compensation.



Anterior OTC/MTL function in hOAs and aMCI individuals. The hyperdifferentiation of semantic representations in ATL is consistent with the results of an fMRI study of recognition memory and aging,¹⁰² in which we found that, compared to YAs, hOAs showed reduced memory-related activity in posterior MTL (hippocampus) but enhanced memory-related activity in anterior MTL (perirhinal cortex—PRC, **Fig. 6-A**). Interestingly, PRC activity predicted accuracy only in hOAs, suggesting that hOAs compensated for hippocampal deficits by overrecruiting PRC. In contrast with hOAs, PRC function is impaired in individuals with higher genetic risk for AD and aMCI.⁵⁹ As for aMCI, an event-related fMRI study with aMCI from co-investigator Kelly Giovanello¹⁰³ found that, compared to hOAs, memory-related activity in PRC and ATL was reduced in aMCI (**Fig. 6-B**). In sum, there is evidence that anterior OTC/MTL functions are impaired in aMCI but are spared in hOAs, who may even over-recruit these regions in some conditions (hyperdifferentiation and increased univariate activity).

The finding that perirhinal cortex (PRC) function is spared in hOAs (**Fig. 6-A**) but impaired in aMCI individuals (**Fig. 6-B**) fits with evidence that PRC atrophy is minimal in hOAs⁴⁵ but substantial in aMCI individuals.²⁹⁻³³ PRC atrophy in aMCI is clinically significant because it correlates with aMCI severity²⁹⁻³³ and predicts conversion to AD.^{30-33,104,105} This is not surprising given that PRC is the first region in the brain to display AD-related *neurofibrillary tangles* (NFTs). NFTs accumulate in medial PRC (*transentorhinal cortex*) during preclinical AD, and then spread to the hippocampus and other regions by the time of the dementia diagnosis (**Fig. 6-C**).¹⁰⁶⁻¹⁰⁸ Given that PRC function is impaired in aMCI but not in hOAs, this region is an ideal target for early AD

108 Given that PRC function is impaired in aMCI but not in hOAs, this region is an ideal target for early AD

detection. The RSA and ERS measures investigated in proposed studies have the potential of being more sensitive to PRC dysfunction than traditional univariate activity measures, and hence, to be useful in early AD diagnosis.

Summary of Specific Aim 1. Our Specific Aim 1 is to *investigate the effects of healthy aging and aMCI on visual vs. semantic features of memory representations*. Visual objects are processed along OTC, with posterior OTC analyzing and storing *visual features* and anterior OTC, mainly *semantic features*. Visual and semantic features are also stored in posterior vs. anterior MTL, respectively. hOAs show *de-differentiation* in posterior OTC,⁹⁻¹⁴ but we have found that they show *hyperdifferentiation* and higher ERS in anterior OTC,⁹⁷ and overrecruit PRC.¹⁰² In contrast, aMCI display deficits in anterior OTC/MTL anatomy and function, consistent with AD pathogenesis. In Aim 1, we test the hypothesis that *visual features in posterior OTC/MTL are impaired in both aMCI and hOAs, whereas semantic features in anterior OTC/MTL are impaired in aMCI but not in hOAs (Hypothesis 1)*. Such dissociation would help distinguish between healthy and pathological aging.

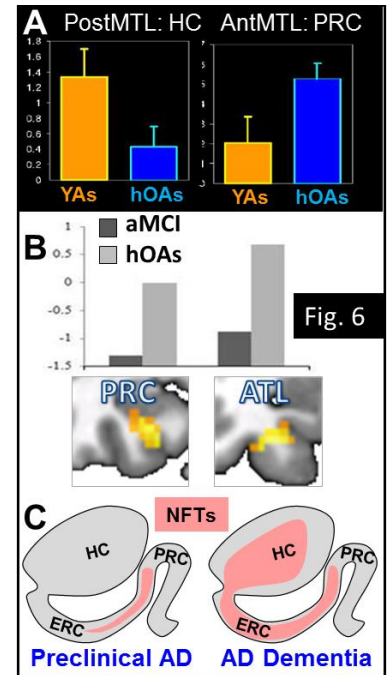
3. Aim 2: Effects of aging and aMCI on the semantic support of memory representations

Semantization of cognition. Cognitive control processes (aka. *fluid intelligence*) decline almost linearly with healthy aging, while semantics (aka. *crystallized intelligence*) continue to accumulate and are preserved in old age.⁶⁶⁻⁶⁹ For example, hOAs retain and use knowledge acquired in school, continue increasing their vocabulary, and remember more facts about the world than YAs.⁶⁹ The decline of fluid intelligence coupled with the preservation of crystallized intelligence leads to an age-related shift from cognitive control to semantics, known as *semantization of cognition*.⁶⁸ This shift impacts a variety of mental abilities, including socio-emotional functioning,¹⁰⁹ decision making,¹¹⁰ and memory.⁶⁹ hOAs overreliance on semantics has both disadvantages and advantages for episodic memory. An example of a disadvantage is an increase in false memory for events consistent with preexistent knowledge,¹¹¹ and an example of an advantage is the *semantic support effect*.

Semantic support effect. The *semantic support effect* refers to the reliable finding that conditions that allow and/or encourage the use of preexistent semantic knowledge tend to attenuate age-related cognitive deficits, including memory deficits.^{23,34-36} In one study,¹¹² for example, hOAs (vs. YAs) were impaired in remembering the layout of a house when it was labeled “building” but not when it was labeled “house,” which encouraged hOAs to tap their knowledge of typical house layouts to help them encode the novel visual stimulus. Given that it enhances memory, it is not surprising that the semantic support effect is used in memory training programs in hOAs and aMCI individuals.^{64,65} However, the semantic support effect is less effective in aMCI individuals, who show a reduced (and sometimes nonsignificant) semantic support effect in recall^{37,38} and recognition,³⁹ consistent with their deficits in semantic memory⁵⁹⁻⁶³ and anterior OTC/MTL function.²⁹⁻³³ When semantic memory is preserved in aMCI individuals, it is associated with a slower conversion to AD,¹¹³ suggesting it operates as a “cognitive reserve”¹¹⁴ that can offset episodic memory deficits. Because the semantic support is attenuated in aMCI individuals, memory performance differences between hOAs and aMCI individuals become more pronounced under supportive conditions. For example, one study found that after passive reading, word recall was similar in aMCI individuals and hOAs, whereas after a deep encoding task, which boosted encoding in hOAs more than aMCI individuals, aMCI individuals were impaired compared to hOAs.³⁸ Thus, investigating the neural mechanisms of semantic support is important not only because it can enhance memory but also because it helps distinguish between healthy and pathological memory deficits.

Varieties of semantic support. We investigate 3 different forms of the semantic support effect, deep encoding, schema consistency, and conceptual expectations, in Studies 1-3, respectively. In **deep encoding** (Study 1), learning is facilitated by an encoding task that encourages participants to process the meaning of incoming information. **Schema consistency** (Study 2) refers to enhanced learning of novel associations when the associations are consistent with preexistent schema. Finally, **valid conceptual expectations** (Study 3) refer to enhanced encoding when participants have accurate predictions about the semantic nature of incoming information. Although these 3 forms of semantic support are related to each other, they involve different cognitive and neural mechanisms. For example, the effects of deep encoding and schema consistency occur at the time the stimulus is encountered, whereas the effects of conceptual expectations start even before the occurrence of the stimulus. Also, while deep encoding and conceptual expectations enhance both item and associative memory, schema consistency applies mainly to associative memory. Moreover, as described in greater detail in the introduction to each study, the neural mechanisms of these three forms semantic support are overlapping but partly different. For example, the memory-enhancing effect of deep encoding has been linked to ventrolateral PFC,^{115,116} whereas the one of schematic consistency have been associated with medial PFC.¹¹⁷⁻¹¹⁹

Summary of Aim 2. Our Aim 2 is to *examine the effects of semantic support on memory representations*



in hOAs and aMCI. The *semantic support effect* refers to the finding that conditions that promote the use of preexistent semantic knowledge tend to attenuate age-related cognitive deficits.^{23,34-36} This effect is weaker in aMCI individuals,³⁷⁻³⁹ consistent with their semantic memory deficits²⁶⁻²⁸ and anterior OTC/MTL dysfunction.²⁹⁻³³ In the Studies 1-3, we respectively investigate 3 different forms of the semantic support effect mediated by overlapping but partly different neural mechanisms: deep encoding, schema consistency, and conceptual expectations. In all cases, we hypothesize that the *semantic support effect enhances the quality of memory representations and subsequent memory to a greater extent in hOAs than in aMCI individuals (Hypothesis 2)*. Given that the semantic support effect is used for memory training in hOAs and aMCI individuals^{59,60} and amplifies memory differences between them,³⁸ understanding the neural mechanisms of this effect is a highly significant goal.

4. Basic methods shared across studies

Studies timeline. The 3 studies are staggered over time (Study 1: Y1-3; Study 2: Y2-4, and Study 3: Y3-5) with testing ending by mid Y5, to allow time for the analyses combined data (see *Combined Analyses* section).

Participants. Each study includes 50 YAs, 50 hOAs, and 50 aMCI. YAs are 18-30 y.o., while hOAs and aMCI individuals are 65-85 y.o. Each participant is scanned in a single experiment, but aMCI individuals will be **C1.8** invited to return in Year 5 for a second neuropsychological assessment, to be used in exploratory analyses. YAs and hOAs are recruited by advertisement and from participant pools. aMCI individuals are recruited by **co-investigators Drs. Johnson and Dr. O'Brien** from their respective memory clinics or using a patient recruitment protocol at Duke (with the diagnosed confirmed by Drs. Johnson or O'Brien). For more details see *Recruitment and Retention Plan* document.

Stimuli. In all studies, the stimuli are pictures of objects from different semantic categories (fruits, tools, etc.) selected from the CabezaLab 1000+ object database, which has been normed for *visual features* (e.g., HMAX, image energy, RGB, Gabor jet, etc.), *semantic features* (e.g., semantic properties, distinctiveness, manipulability, etc.), *categorical relationships* (category name associations, typicality, etc.), and *name properties* (name agreement, frequency, etc.). The objects used in the studies are all common (easy to name) and with high name agreement. In all pictures, the objects are 7.5° in size, have similar resolution, brightness, etc. It is worth noting that we do not use audition as a control because the representations of “auditory objects” (e.g., a **C1.8** drill’s sound) are (a) very different and difficult to the ones of visual objects, and (b) likely to be affected by aging and aMCI as much as, or more than visual object representations. **Memory paradigm.** In all studies, the memory paradigm consists of 3 phases (see the design of Study 1 in **Fig. 7** as an example). **(1) During the scanned encoding task**, participants view object pictures under specific processing instructions, which vary across studies. In all studies, the main experimental manipulation occurs during encoding. **(2) During the scanned word task**, the names of studied objects are presented and participants rate their memory from 1 (cannot recall the object) to 4 (excellent memory). As noted, this kind of imagery-based object recall task involves the retrieval of both semantic and visual feature representations. **(3) Finally, during the non-scanned picture test** (forced-choice recognition), the picture of each encoded object is mixed with 3 new exemplars of the same object (same name), and participants select the studied exemplar, rating their confidence. The picture test is not scanned because 4 objects on the screen preclude the identification of object-specific memory representations. However, behavior in this test is used to assess visual features representations during encoding, which will then be used to predict performance in the picture memory test, and during the scanned word test, which are associated with picture test performance. It is worth noting that testing memory for the same objects twice is not a problem because remembering one encountered a type of object (e.g., a hammock) does not imply one can identify the specific picture seen in a forced-choice recognition test (e.g., a particular hammock picture). In fact, as shown by Bahrick and Boucher’s^{78,79} studies, and replicated in our own studies, memory performance in these two tasks, when administered in this order, is uncorrelated.

Shared analyses. RSA. As in our preliminary RSA study (**Fig. 2**), we use a *visual RDM* coding for visual similarity (e.g., line orientation, spatial frequency) and a *semantic RDM* coding for semantic similarity (from association norms), and then relate each of these matrices to the *brain similarity matrix* to obtain a *RDM-brain fit* measure in every ROI. The visual RDM-brain fit assesses visual feature representations, and the semantic RDM-brain fit, semantic feature representations. We then identify regions where the visual RDM-brain fit and/or the semantic RDM-brain fit predicts subsequent memory performance in the “hybrid” imagery-based *word task* and/or the *picture test*. **ERS.** As in our preliminary ERS study (**Fig. 3**), we calculate *item-* and *set-level ERS* (matching vs. mismatching study-test pairs) and link them to memory performance in the word and picture tests. **Healthy aging and aMCI effects.** We measure healthy aging effects by comparing **[hOAs vs YAs]** and aMCI effects by comparing **[aMCI individuals vs hOAs]**. We then compare these two contrasts to identify qualitative differences between the effects healthy aging and aMCI.

General hypotheses. All studies test 2 main hypotheses. We also test several alternative hypotheses but, for sake of space, we mention here only two of them. **Hypothesis 1** postulates that *visual feature representations in posterior OTC/MTL are impaired in both aMCI and hOAs, whereas semantic feature representations*

in anterior OTC/MTL are impaired in aMCI but not in hOAs. An alternative hypothesis is that semantic feature representations are spared in both hOAs and aMCI (**Alternative Hypothesis 1**). Confirming Hypothesis 1 would indicate that aMCI individuals have a fundamental deficit in semantic feature representations, while confirming its alternative would suggest that semantic deficits in aMCI²⁶⁻²⁸ reflect other mechanisms, such as downstream semantic processing deficits, perhaps involving VLPFC. **Hypothesis 2** states that *semantic support enhances the quality of memory representations and subsequent memory to a greater extent in hOAs than in aMCI individuals*. An alternative hypothesis is that the enhancement is similar in both groups (**Alternative Hypothesis 2**). Confirming Hypothesis 2 would cast doubt on using semantic support for memory training in aMCI individuals, while confirming its alternative would support such an approach. Although these and other alternative hypotheses are examined in every study, given space limitations, they are not mentioned further in this proposal.

Testing the two main hypotheses in all 3 studies is useful for several reasons. **First**, it ensures the replicability and generalizability of the main findings. For example, our finding of spared semantic feature representations in hOAs (**Fig. 5-A**) could be limited to the particular stimuli (scenes) and encoding task (representativeness ratings) used in this study. If Studies 1-3 replicate this finding using different stimuli (clear/blurred objects, schema-congruent vs incongruent picture-word pairs, etc.) and encoding tasks (deep vs. shallow, relatedness ratings, and object-pseudo-object judgments), the reliability and robustness of this finding would be confirmed. **Second**, each study investigates a different factor contributing to the deficit in visual feature representation in hOAs and aMCI individuals (visual input, relational binding, and top-down attention), and hence, it generates converging evidence of different aspects of this deficit. **Finally**, each study examines a different form of the semantic support effect (deep encoding, schema consistency, and conceptual expectations), thereby providing a more complete picture of the effects of aging and aMCI on this important phenomenon.

5. Studies

Study 1

Introduction. Regarding **Aim 1**, Study 1 tests **Hypothesis 1** and examines one factor contributing to impaired visual feature representations in hOAs and aMCI individuals: a **visual input deficit**. As noted before, in both healthy aging and AD, perceptual and cognitive deficits are strongly correlated.^{3,4,120} A recent review by **co-investigator David Madden**¹²¹ concluded the hypothesis that in hOAs, the decline of the peripheral visual system leads to a visual input deficit that impairs cognition is supported by behavioral evidence. For example, this hypothesis is consistent with evidence that degrading visual stimuli yields memory deficits in YAs that resemble those in OAs.¹²²⁻¹²⁴ Yet, the neural mechanisms linking the visual input deficit to memory impairments are unclear. Study 1 investigates this issue by comparing blurred vs. clear stimuli.

Regarding **Aim 2**, Study 1 tests **Hypothesis 2** focusing on a 1st form of the semantic support effect: **deep encoding**. In general, *deep encoding* tasks that encourage participants to process the meaning of stimuli (e.g., living/nonliving judgments) yield better retention than *shallow encoding* tasks that encourage them to process physical aspects of the stimuli (e.g., lowercase/uppercase judgments).^{125,126} In YAs, deep encoding does not often surpass *intentional encoding*, because YAs tend to spontaneously use deep encoding strategies when trying to memorize information. In contrast, hOAs rarely use self-initiated encoding strategies, and hence, they tend to benefit more than YAs by deep encoding tasks.¹¹⁵ In one study, for example, name recognition was worse in hOAs than YAs following intentional encoding (55% vs 69%) but it was slightly better in hOAs than YAs after deep encoding (75% vs 73%).¹²⁷ Compared to hOAs, the beneficial effect of deep encoding is smaller in aMCI,^{37,38} consistent with their semantic memory deficits^{59,60} and anterior OTC/MTL dysfunction.²⁹⁻³³ For this reason, deep encoding, like other forms of semantic support, tends to amplify memory differences between aMCI individuals and hOAs. For instance, one study³⁸ found that word recall was similar in aMCI individuals and hOAs after word reading but it was significantly worse in aMCI individuals than hOAs after deep encoding because aMCI individuals did not benefit as much by deep encoding (~15% improvement) as hOAs did (~50% improvement). Thus, comparing hOAs vs. aMCI individuals under deep encoding reveals the limits of aMCI individuals' memory abilities, facilitating the distinction between healthy and pathological memory decline.

To our knowledge, only two fMRI studies have investigated deep encoding in hOAs,^{128,129} but one of them (from our lab) focused on emotion effects¹²⁸ and the other did not include YAs and could not measure age effects.¹²⁹ Deep encoding in aMCI was examined in one study³⁹ but it used a blocked design that precluded direct activity-performance links. Also, none of these studies with hOAs or aMCI individuals examined memory representations.

Methods. During encoding and retrieval, half of the objects are **clear** and half are **blurred** (**Fig. 7**), and they are tested in the same format during retrieval. During encoding, each fMRI run consists of a **deep encoding block**, in which participants make living/nonliving decisions, and a **shallow encoding block**, in which they make clear/blurred decisions. The order of these two blocks is counterbalanced across runs and participants. The word and picture tests were described in Section 4.

Aim 1 predictions. Study 1 tests **Hypothesis 1** (visual feature representations are impaired in both aMCI

and hOAs while semantic feature representations are impaired only in aMCI), and examines the contribution of visual input degradation to visual feature representation deficits in hOAs and aMCI individuals. We predict that, compared to clear objects, blurred objects will worsen the visual feature representation deficit in posterior OTC/MTL in hOAs, and that this effect will be more pronounced in aMCI individuals. Also, we predict that blurred objects will be able to mimic posterior OTC/MTL dedifferentiation in YAs, supporting the idea that age-related dedifferentiation is partly due to input loss.

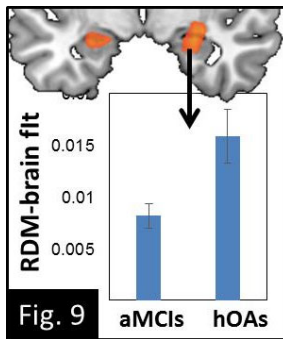
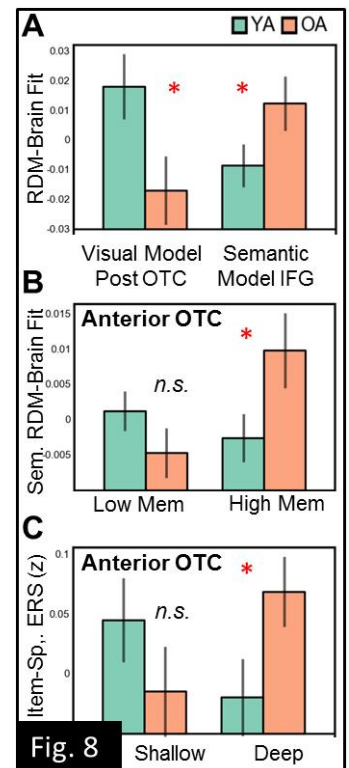
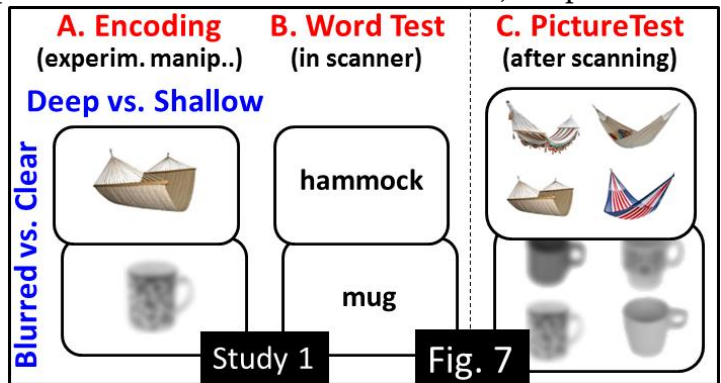
Aim 2 predictions. In Study 1, we test **Hypothesis 2** by comparing deep vs. shallow encoding. We predict that, *compared to shallow encoding, deep encoding will enhance the quality of memory representations and subsequent memory to a greater extent in hOAs than in aMCI individuals.* We expect that deep encoding will enhance semantic feature representations in anterior OTC/MTL. Also, we expect that deep encoding will be associated with increased functional connectivity with between anterior OTC/MTL and left ventrolateral PFC, a region strongly associated with semantic processing.⁷⁰ Given that shallow encoding promotes the encoding of visual features of the stimuli, an interesting exploratory question is whether this task will attenuates the visual feature representations deficit in posterior OTC/MTL in hOAs and aMCI individuals.

Study 1 pilot. We conducted an fMRI pilot study 18 YAs, 16hOAs comparing deep vs. shallow tasks. Memory performance in the post-scan picture test was good in both YAs ($M=0.72$, above chance: $p<0.0001$) and hOAs ($M=0.71$, above chance: $p<0.0001$). RSA results replicated our previous finding of differential age effects on visual vs. semantic feature representations (see **Fig. 5-A**). As illustrated by **Fig. 8-A**, we found that visual RDM-brain fit in posterior OTC was reduced in hOAs whereas the semantic RDM-brain fit in anterior brain regions (VLPFC in this case) was enhanced by aging (Age Group x Memory interaction: $p<0.01$). Also consistent with our previous study, hOAs' reliance on semantic feature representations in anterior OTC was associated with better memory performance (**Fig. 8-B**), supporting a compensation account. Finally, in keeping with the semantic support prediction, the deep encoding task enhanced ERS in anterior OTC in hOAs (**Fig. 8-C**). Although these are preliminary analyses of a sample that is less than half the one proposed, they support the feasibility of the proposed design for detecting the effects of age, deep encoding, and their interaction on OTC memory representations.

Study 2

Introduction. Regarding **Aim 1**, Study 1 tests **Hypothesis 1** and additionally examines another factor contributing to impaired visual feature representations in hOAs and aMCI individuals: **associative memory deficit**. Associations (e.g., the hammock has red strips like the US flag) facilitate the encoding of visual features (e.g. the US-flag shade of red),¹³⁰ possibly through feedback connections from MTL to posterior OTC. hOAs are more impaired associative than in item memory.^{36,131} This associative memory deficit has been linked to MTL dysfunction,¹³² which is more severe in aMCI individuals than in hOAs.^{133,134} For example, in an aMCI study from study consultant Kelly Giovanello,¹⁰³ posterior MTL activity was greater for associative than item memory in hOAs and this activity was reduced in aMCI individuals. To examine aMCI effects on associative memory representations, we re-analyzed this dataset using RSA, with an RDM comparing associative vs. item trials. Using a searchlight approach, we found that the RDM-brain fit was weaker in aMCI individuals than hOAs ($p < 0.01$) in several MTL regions, including the hippocampus (**Fig. 9**). These results support the idea that the quality of associative memory representations in MTL is reduced in aMCI. In Study 2, we will use a similar approach to assess the associative memory representation deficit in aMCI but will additionally link this deficit in picture test performance to examine how the associative memory deficits impair visual feature representations.

Regarding **Aim 2**, Study 1 tests **Hypothesis 2** focusing on a 2nd form of the semantic support effect: **schema consistency**. hOAs associative memory deficit is markedly reduced when the new associations are consistent with preexistent knowledge.⁶⁹ For example, a study found that word-pair cued recall deficits in hOAs were large for unrelated word-pairs, modest for weakly related word-pairs, and negligible for strongly related



word-pairs²³ (see also ^{24,25,36,135-138}). Another study¹³⁹ found that, compared to YAs, hOAs were impaired in remembering the prices of grocery items when the prices were unrealistic but not when they were realistic (i.e. consistent with preexistent schemas). Similar to the effect of deep encoding, hOAs often benefit more than YAs by schema consistency. For instance, in one study in which memory for the speakers of various statements was enhanced by schema consistency (e.g., the statement about a medicine was uttered by the doctor), the improvement was almost 3 times as large in hOAs than in YAs.¹⁴⁰ As in the case of deep encoding, the benefit of schema consistency is reduced in aMCI individuals. For example, a study found that hOAs showed better associative memory for faces paired with congruent information (e.g., profession or name) but the effect significantly attenuated in aMCI individuals,¹⁴¹ thereby increasing the difference between hOAs and aMCI individuals. In fact, hOAs showed better memory than aMCI individuals congruent pairs but not for incongruent pairs (see also ¹⁴²). Thus, as in the case of deep encoding, schema consistency amplifies differences between hOAs and aMCI individuals, and should facilitate the distinction between these two groups.

The enhancing effect of schema consistency on associative memory has been linked to *medial prefrontal cortex (mPFC)*. This region is assumed to integrate new information into networks of pre-existing neocortical associations,^{143,144} complementing the hippocampus:¹¹⁷⁻¹¹⁹ The hippocampus is assumed to mediate information binding regardless of whether is congruent or incongruent with preexisting knowledge, whereas mPFC mediates primarily schema-congruent binding. In a preliminary *schema fMRI study*,¹⁴⁵ participants encoded scenes paired with object names, which were either *schema-congruent* (e.g., a graduation scene paired with word “diploma”) or *schema-incongruent* (**schema-inconsistent**, e.g., a bedroom scene paired with word “saw”). The study yielded 3 main findings. **(1)** As illustrated by **Fig. 10-A**, bilateral hippocampal activity (in blue in **Fig. 10-B**) predicted subsequent memory (1 to 4) similarly for schema-congruent and -incongruent pairs, whereas mPFC activity (in red in **Fig. 10-B**) predicted subsequent memory for schema-congruent but not schema-inconsistent pairs. **(2)** Item-specific ERS (in green in **Fig. 10-B**) was found in OTC and posterior cingulate for both schema-congruent and incongruent pairs, suggesting shared visual memory representations. **(3)** The impact of OTC reactivation on memory performance (**Fig. 10-C**) was mediated by the hippocampus (blue line) for *both* schema-consistent and -inconsistent pairs but by mPFC (red line) *only* for schema-consistent pairs. We assume that the schema consistency effect in hOAs is partly mediated by mPFC. In contrast with the hippocampus, which is one of the regions most affected by aging,¹⁴⁶ mPFC shows relatively small age-related anatomical decline.¹⁴⁷⁻¹⁴⁹ Moreover, in a *fMRI study on memory and decision making*¹⁵⁰ we found increased mPFC activity in hOAs than YAs, which associated with faster retrieval RTs in hOAs, consistent with compensation.

Methods. In each encoding trial, participants view the *picture of an object* and listen to *one word* (**Fig. 11**). 3 types of encoding trials are randomly intermixed: schema-consistent, schema-inconsistent, and item trials. In schema-consistent and -inconsistent trials, the words reflect the name of a scene. In **schema-consistent trials**, the scene often contains the object, whereas in the **schema-inconsistent trials**, the scene is unrelated to the object. In **item trials**, the word is an object name that either matches (80% trials) or mismatches (20% of trials) the picture. In all 3 types of trials, participants respond if the picture and word match or mismatch; they answer “match” for schema-consistent and matching item trials, and “mismatch” or schema-inconsistent and mismatching item trials. The latter are used as “catch trials” and are not analyzed. Word and picture tests were described in Section 4.

Aim 1 predictions. While testing **Hypothesis 1** (visual feature representations are impaired in both aMCI and hOAs; semantic feature representations, only in aMCI), Study 2 investigates the impact of impaired associative memory representations to visual feature representations in hOAs and aMCI individuals. We assess the quality of associative memory representations with RSA based on RDMs comparing relational trials (both schema-consistent and inconsistent) to item trials. We predict that impaired associative representations in the hippocampus will be associated both degraded visual representations in posterior OTC and with poor subsequent picture memory. We expect that this effect will be more pronounced in aMCI than in hOAs.

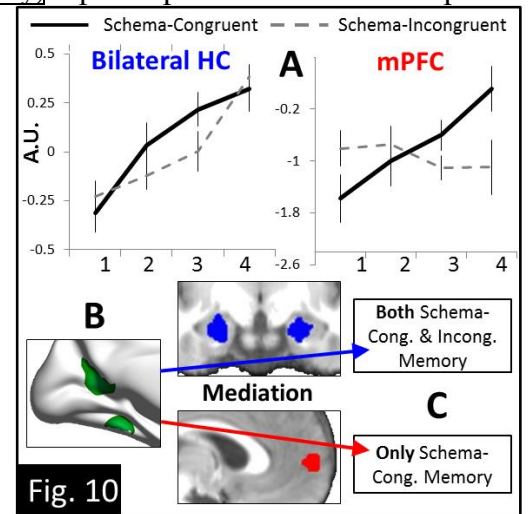


Fig. 10

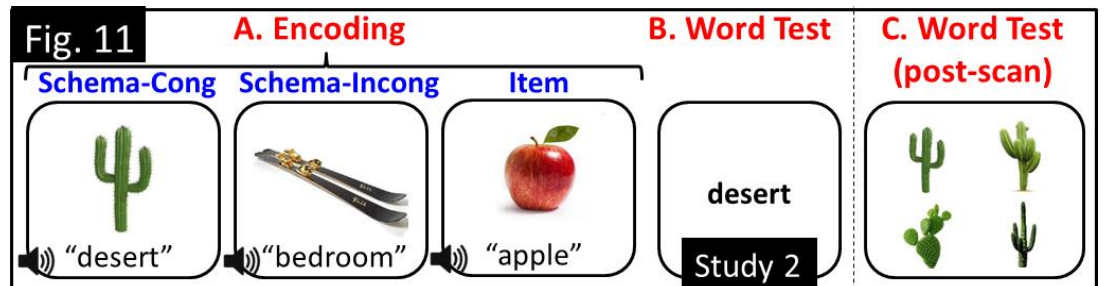


Fig. 11

Aim 2 predictions. In Study 2, **Hypothesis 2** is tested by comparing schema-consistent vs. -inconsistent pairs. On the basis of our preliminary schema study (see **Fig. 10-C**), we predict that *compared to schema-inconsistent pairs, schema-consistent pairs will augment the impact of memory representations on subsequent memory via mPFC mediation, and that this effect will be weaker in aMCI individuals than in hOAs*. If mPFC is partly compensating for hippocampal deficits in hOAs, then the role of mPFC will be negatively correlated with the quality of associative memory representations in the hippocampus. We assume that compensatory mPFC recruitment will be attenuated in aMCI.

Study 3

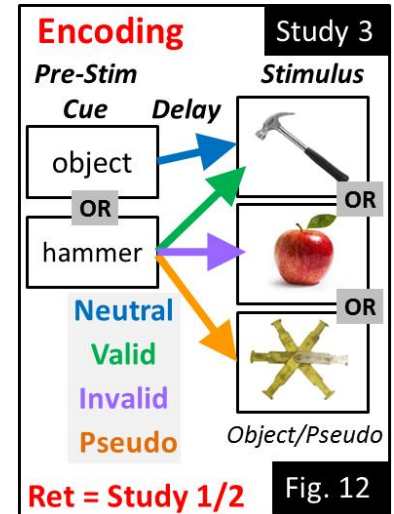
Introduction. Regarding **Aim 1**, while testing **Hypothesis 1**, Study 3 examines a third factor potentially contributing to impaired visual feature representations in hOAs and aMCI individuals: **attenuated top-down attention**. It is well known that PFC-mediated top-down attention enhances visual processing by upregulating OTC function.^{151,152} For example, in the absence of visual stimuli, OTC regions specialized in processing faces (*fusiform face area*—FFA) and places (*parahippocampal place area*—PPA) are activated when participants imagine^{153,154} or prepare to detect or encode^{155,156} faces or places, respectively. These mechanisms are impaired in hOAs, who—for example—fail to activate the FFA when preparing to encode faces¹⁵⁷ and suppress PPA activity when scenes must be ignored.¹⁵⁸ These impairments have been attributed to a deficit in PFC-mediated modulation of OTC, possibly reflecting white-matter deficits in the tract linking PFC and OTC: the *inferior fronto-occipital fasciculus* (IFOF). For example, a study found that hOAs' errors in difficult face-discrimination trials correlated with white-matter deficits in the IFOF.¹⁵⁹ Also, a reduction in PFC-mediated modulation could partly account for age-related dedifferentiation of OTC representations. Consistent with this idea, a study found that disrupting PFC using transcranial magnetic stimulation (TMS) mimicked OTC dedifferentiation in YAs.¹⁶⁰ Study 3 investigates the role of top-down attention deficits in the impairment of visual feature representations in hOAs and aMCI individuals by measuring PFC-OTC functional connectivity before each object is presented.

Regarding **Aim 2**, Study 1 tests **Hypothesis 2** focusing on a 3rd form of the semantic support effect: **conceptual expectations**. Most fMRI studies on top-down attention effects on visual processing have focused on spatial attention or featural attention, and very little is known regarding the role *conceptual expectations*. One exception is Cristescu and Nobre's (2006) fMRI study,¹⁶¹ in which a *pre-stimulus cue*, the word "living" or "nonliving", generated valid or invalid expectations for the upcoming word *stimulus* (e.g., "dog"). This task recruited dorsal fronto-parietal regions shared with spatial attention, as well as semantics-related regions, such as left ventrolateral PFC. **Study 2** uses a similar paradigm to Cristescu and Nobre but it (1) investigates not only visual processing but also on subsequent memory; (2) measures not only univariate activity but also memory representation quality; and (3) examines not only YAs but also the effects of healthy aging and aMCI.

Method. **Fig. 12** shows the design of Study 3. During **encoding**, trials have 3 phases: pre-stimulus cue, delay, and stimulus. The *pre-stimulus cue* is either the uninformative word "object" or a word indicating the likely identity of the stimulus. The *delay* is jittered to allow separate fMRI measures of the pre-stimulus cue and the stimulus. During the *stimulus*, participants make object/pseudo-object decisions. Pseudo-objects are unfamiliar objects of unknown name and function. There are 4 types of trials with only the first 3 used in the analyses: (1) **neutral trials** (uninformative pre-stimulus cue: "object"), (2) **valid trials** (65%), (3) **invalid trials** (25%), and (4) pseudo-objects (5%; not analyzed).

We tested the paradigm of Study 3 in a small (N=9) **behavioral pilot study**. Compared to neutral trials, the RTs of object/pseudo-object decisions were faster for valid trials and slower for invalid trials (**Fig. 13-A**), confirming the effectiveness of the validity manipulation. Interestingly, subsequent recognition memory (Hits-FAs) was better for *both* valid and invalid trials compared to neutral trials (**Fig. 13-B**). Thus, encoding was enhanced not only by valid expectations but also by invalid expectations, consistent with the memory-enhance effect of novelty^{162,163} and prediction error,^{164,165} respectively.

Aim 1 predictions. While testing general **Hypothesis 1** (visual feature representations are impaired in both aMCI and hOAs, but semantic feature representations, will only be impaired in aMCI), Study 3 examines the contribution of attenuated top-down modulation to deficits in visual feature representations in hOAs and aMCI individuals. We predict that (1) delay-period functional connectivity between PFC and OTC will be greater for informative than neutral trials reflecting top-down attention, (2) this connectivity will be attenuated in hOAs and aMCI individuals, and (3) reduced connectivity will correlate with visual feature representation deficits in posterior OTC in these groups.



Aim 2 predictions. In Study 3, **Hypothesis 2** (greater semantic support effect for hOAs than aMCI individuals) is tested by comparing valid and neutral pairs: *compared to neutral trials, valid trials enhance the quality of semantic feature representations in anterior OTC/MTL and subsequent memory in the word task to a greater extent in hOAs than in aMCI*. In particular, we predict that that greater connectivity between PFC and anterior OTC/MTL for valid than neutral trials will correlate with semantic feature representation quality (semantic RDM-brain fit) in anterior OTC and subsequent performance in the word memory task, and these effects will be reduced in aMCI compared to hOAs.

Exploratory question. In addition to Hypotheses 1 and 2, Study 3 investigates the neural mechanism of the finding that invalid trials yield better subsequent memory than neutral trials (**Fig. 13-B**). This finding may reflect a memory-enhancement effect of novelty or prediction error for unexpected objects, or more elaborative processing (supported by longer encoding RTs, **Fig. 13-A**). Given that conceptual expectations were violated, the additional processing is likely to be perceptual in nature. If a hammer is expected but an apple appears (see example in **Fig. 12**), additional visual analyses are required to offset the incorrect perceptual expectations. Thus, unlike the valid > neutral effect, which we assume reflect enhanced semantic feature representations, we predict that the invalid > neutral effect reflects enhanced visual feature representations. If so, this effect could attenuate deficits in these representations in hOAs and aMCI individuals.

Exploratory analyses

By mid Year 5, all participants of Studies 1-2 and most participants of Study 3 will have been scanned, and hence, we will be able to conduct analyses combining data from multiple studies. Given that scanned **C1.11** participants are different in each study, the combined analyses could include as many 120 YAs, 120 hOAs, and 120 aMCI individuals. Although experimental conditions differ across studies, the deep-clear condition of Study 1, the item condition of Study 2, and the neutral condition of Study 3 are relatively similar, and could be combined. It is worth noting that RSA is an excellent method for combining different types of data, such as electrophysiological and fMRI data in humans or human fMRI with monkey electrophysiology.¹⁶⁶⁻¹⁶⁸ We will also explore factors difficult to examine in individual studies, such as gender effects.

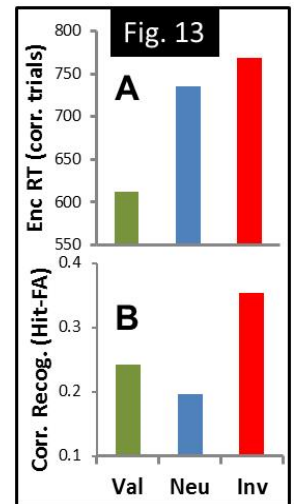
We will also explore the possibility of using RSA from Studies 1-2 to predict cognitive decline several years later. hOAs and aMCI individuals from Studies 1-2 will be invited to return to the lab in Year 5 (2.5 yrs later on average) for follow-up neuropsychological tests (no MRI). Small learning effects may occur in these tests **C1.11** but they will not be a serious problem because will not compare individuals from different groups (only hOAs to hOAs or aMCI individuals to aMCI individuals). In addition to decline in both groups, we will also explore the possibility of predicting conversion to aMCI in hOAs and to AD in aMCI individuals. As described in the *Recruitment and Retention plan*, if conversion is confirmed by Dr. Johnson or Dr. O'Brien the participant will **C2.3** be classified as "converter to aMCI" or "converter to AD." We will then conduct separate logistic regression analyses with conversion to aMCI or conversion to AD as dependent variable and RSA and other variables (memory performance) as regressors. As noted before, there is evidence that behavioral measures of visual memory can predict AD more than a decade before diagnosis⁶ and with high specificity.⁵⁸ We will explore if RSA measures can surpass behavioral memory measures. Although the number of converters is likely to be small (e.g., ~30 converters to AD if all 100 aMCI individuals from Studies 1-2 return for follow-up neuropsychological assessment and they convert at an estimated rate of 15% per year¹⁶⁹), the results could support a future grant application.

Finally, in addition to the regions-of-interest in our model of object memory (**Fig. 1**) we will also explore the role of other brain regions. In particular, we will use the Power atlas (Power et al., Neuron, 2011) and **C1.6** examine components of the "Memory Retrieval" and default networks relevant to our hypotheses, such as **C1.7** the angular gyrus (AG). Regarding Hypothesis 1, AG has been associated with semantic processing (Binder et al., 2009, Cereb Ctx) and hence, we expect this region to behave similarly to ATL and show impaired semantic representations in aMCI individuals but not in hOA. Regarding Hypothesis 2, we expect this region to show enhanced semantic representations in response to semantic support in hOAs but not in aMCI individuals. We will also explore other components of the memory retrieval and default networks.

6. Methodological details

Sessions. Each study consists of 2 sessions. During **Session 1** (~3 hrs with breaks), participants (1) fill out several questionnaires (MRI safety, etc.), (2) perform various visual function tests (near and distant acuity, contrast sensitivity, etc.), which are used as co-variates in some fMRI analyses, and (3) complete a comprehensive neuropsychological battery (BDI, MoCA, and NACC-UDS v3, see *Recruitment and Retention Plan* document), and (4) practice the fMRI tasks. During **Session 2**, participants undergo structural MRI, DTI, and fMRI of brain activity during object encoding and retrieval.

MRI session. In all studies, the 120-min MRI session in BIAC's GE 3T Premier UHP scanner is employed.



which is capable of both multiband functional and structural imaging. The 2-hr MRI session includes (1) structural MRI, (2) DTI, and (3) multiple fMRI runs. (1) *Structural MRI*. T1-weighted SPGR scans are acquired with a 450-ms TR and 1-mm³ isotropic voxels (5 min). (2) *DTI*. 60 directions are acquired with a 1.7-s TR and 1-mm³ isotropic voxels (8 min). (3) *fMRI runs*. A BOLD-sensitive inverse spiral sequence is collected from 126 contiguous slices (same slice prescription as structurals) with a 2-s TR, and 2-mm³ isotropic voxels. The fMRI scans include (i) *lateral occipital complex* (LOC) localizer (object vs. scrambled objects), (ii) a resting-state fMRI, (iii) multiple encoding and retrieval fMRI runs, and (iv) cerebrovascular reactivity (CVR) scan. During this scan, subjects breathe in or out every 3-s, then hold their breath for 20-s, and finally rest for 20-s. This cycle is repeated 7 times for a total of approximately 5-mins. A rise-time analysis is used to estimate both amplitude and rate of change of the hypercapnic response from baseline. C1.9
C1.10

Conservative preprocessing pipeline. To avoid potential confounds in cross-group comparisons, we control for motion (6 motion parameters and their derivatives), include the WM/CSF timeseries regressors, regress out TRs where participants moved a lot, denoise the data, etc. Also, we use DARTEL to create study specific templates, and for regions of interest, we control for number of voxels and SNR.

Representational analyses. Following univariate fMRI analyses using well-established SPM12 methods, similarity measures are calculated each encoding trial beta image E_i from the voxelwise pattern within each anatomical ROI (Harvard-Oxford atlas without cerebellum) or within 3x3x3 voxel searchlight boxes surrounding each voxel¹⁷⁰. Within each ROI or searchlight box, univariate activity across trials is normalized to remove the influence of overall univariate activity in similarity calculations. Similarity is analyzed in 2 ways: model-based RSA and ERS. **RSA.** The pattern vector for E_i is correlated with the pattern vector of each other trial in a task or condition, yielding a *brain similarity matrix* that expresses the similarity of evoked patterns between trials. This matrix is then correlated with one particular representational dissimilarity matrix (**RDM**), and the resulting (2nd-order) **RDM-brain fit** expresses the correspondence between brain-based similarity and a particular model. To examine memory effects, this correspondence is computed separately for subsequently remembered and forgotten trials. **ERS.** The pattern vector for each encoding trial E_i is instead correlated with the vector of the corresponding retrieval trial (*item-level pairs*: E_1-R_1 , E_2-R_2 , etc.) or to each of the other retrieval trials (*set-level pairs*: average of E_1-R_2 , E_1-R_3 , etc.). Greater ERS for item- than set-level pairs indicates sensitivity to memory representations at the level of individual stimuli.

Scientific Rigor and Sex as a Biological Variable

The 4 proposed studies are based on solid scientific premises, as they are supported both by strong prior literature and substantial preliminary evidence. In our **data collection phase**, we will ensure scientific rigor and appropriate sample size. Cognitive psychology^{171,172} and neuroimaging¹⁷³ studies have been plagued by concerns about reproducibility based on inflated false-positive findings. The danger of both false-positives and false-negatives can be greatly reduced by employing sample sizes informed by appropriate effect size estimates¹⁷⁴. Since effect sizes calculated from small studies are likely inflated, we here base our sample sizes on effect size estimates from a recent analysis of 500 fMRI data sets,¹⁷³ which showed that cognitive manipulations of the kind we study here typically have only moderate effect sizes (Cohen's $d \sim 0.5$). For a desired power of 0.8 to detect an effect size of 0.5 in within-subjects contrasts (correlated groups t-test), and assuming a conservative (low) level of correlation between paired observations ($r=0.3$), a sample size of $N=45$ is required (calculation based on GPOWER).¹⁷⁵ Thus, we use $N=45$ as the minimum sample size in the proposed studies. Given that $\sim 10\%$ of participants' data must be excluded from the analyses due to poor behavioral performance, movement in the scanner, etc., 50 participants are included in each group (YAs, hOAs, and aMCI individuals) in every study.

In our **analysis phase**, we will ensure scientific rigor as follows. To address bias in ANOVAs due potential violation of the sphericity assumption, we will estimate sphericity and correct for this bias using a Greenhouse-Geisser correction. To ensure replicability in our findings, we will format all incoming data according to the Brain Imaging Data Structure (BIDS) format to allow for easy sharing of neuroimaging data, and report analytical procedures according to the Committee on Best Practice in Data Analysis and Sharing (COBIDAS) report for fMRI data analysis,¹⁷⁶ and all neuroimaging results will be subject to statistical correction for multiple comparisons using the latest consensus methods.¹⁷⁷ Finally, to ensure generalizability of our findings, we will perform a k-fold cross-validation on all fMRI analyses. In k-fold cross-validation, data is divided into k independent subsets. On each fold, one subset is held out as the test set, and the others used for training. Empirical studies suggest that choices of k in the range of 5 to 10 yield good results in practice.¹⁷⁸

Finally, regarding **sex as a biological variable**, given that we do not investigate memory tasks sensitive to sex differences, such as spatial working memory and emotional memory,¹⁷⁹ we expect these differences will be small and nonsignificant in the YA samples. In contrast, sex can modulate the effects of aging on the brain,¹⁸⁰ including memory mechanisms.¹⁸¹ We do not have specific hypotheses about these effects but we will explore them by comparing male vs female participants. Although we may be underpowered to observe sex differences with 45 hOAs, the outcome of this study could be used to estimate effect sizes to adequately power gender differences in future studies.

Bibliography

- 1 Kirby, E., Bandelow, S. & Hogervorst, E. Visual Impairment in Alzheimer's Disease: A Critical Review. *J. Alzheimers Dis.* **21**, 15-34 (2011).
- 2 Tzekov, R. & Mullan, M. Vision function abnormalities in Alzheimer disease. *Surv. Ophthalmol.* **59**, 414-433, doi:10.1016/j.survophthal.2013.10.002 (2014).
- 3 Lindenberger, U. & Baltes, P. B. Sensory functioning and intelligence in old age: A strong connection. *Psychol. Aging* **9**, 339-355 (1994).
- 4 Baltes, P. B. & Lindenberger, U. Emergence of a powerful connection between sensory and cognitive functions across the adult life span: A new window to the study of cognitive aging? *Psychol. Aging* **12**, 12-21 (1997).
- 5 Lin, M. Y., Gutierrez, P. R., Stone, K. L., Yaffe, K., Ensrud, K. E., Fink, H. A., Sarkisian, C. A., Coleman, A. L. & Mangione, C. M. Vision impairment and combined vision and hearing impairment predict cognitive and functional decline in older women. *J. Am. Geriatr. Soc.* **52**, 1996-2002, doi:10.1111/j.1532-5415.2004.52554.x (2004).
- 6 Kawas, C. H., Corrada, M. M., Brookmeyer, R., Morrison, A., Resnick, S. M., Zonderman, A. B. & Arenberg, D. Visual memory predicts Alzheimer's disease more than a decade before diagnosis. *Neurology* **60**, 1089-1093 (2003).
- 7 Petersen, R. C. Mild cognitive impairment as a diagnostic entity. *J. Intern. Med.* **256**, 183-194 (2004).
- 8 Winblad, B., Palmer, K., Kivipelto, M., Jelic, V., Fratiglioni, L., Wahlund, L. O., Nordberg, A., Bäckman, L., Albert, M. & Almkvist, O. Mild cognitive impairment—beyond controversies, towards a consensus: report of the International Working Group on Mild Cognitive Impairment. *J. Intern. Med.* **256**, 240-246 (2004).
- 9 Park, D. C., Polk, T. A., Park, R., Minear, M., Savage, A. & Smith, M. R. Aging reduces neural specialization in ventral visual cortex. *Proc Natl Acad Sci U S A* **101**, 13091-13095 (2004).
- 10 Voss, M. W., Erickson, K. I., Chaddock, L., Prakash, R. S., Colcombe, S. J., Morris, K. S., Doerksen, S., Hu, L., McAuley, E. & Kramer, A. F. Dedifferentiation in the visual cortex: An fMRI investigation of individual differences in older adults. *Brain Research* **1244**, 121-131 (2008).
- 11 Park, J., Carp, J., Hebrank, A., Park, D. C. & Polk, T. A. Neural Specificity Predicts Fluid Processing Ability in Older Adults. *J. Neurosci.* **30**, 9253-9259 (2011).
- 12 Kalkstein, J., Checksfield, K., Bollinger, J. & Gazzaley, A. Diminished top-down control underlies a visual imagery deficit in normal aging. *J. Neurosci.* (in press).
- 13 Lee, Y., Smith, C. R., Grady, C. L., Hoang, N. & Moscovitch, M. Broadly Tuned Face Representation in Older Adults Assessed by Categorical Perception. *J. Exp. Psychol.-Hum. Percept. Perform.* **40**, 1060-1071, doi:10.1037/a0035710 (2014).
- 14 Park, J., Carp, J., Kennedy, K. M., Rodrigue, K. M., Bischof, G. N., Huang, C. M., Rieck, J. R., Polk, T. A. & Park, D. C. Neural Broadening or Neural Attenuation? Investigating Age-Related Dedifferentiation in the Face Network in a Large Lifespan Sample. *J. Neurosci.* **32**, 2154-2158, doi:10.1523/jneurosci.4494-11.2012 (2012).
- 15 Zacks, R. T., Hasher, L. & Li, K. Z. H. in *Handbook of aging and cognition II* Vol. 293-357 (eds F. I. M. Craik & T. A. Salthouse) (Erlbaum, 2000).
- 16 Nyberg, L., Maitland, S. B., Ronnlund, M., Backman, L., Dixon, R. A., Wahlin, A. & Nilsson, L. G. Selective adult age differences in an age-invariant multifactor model of declarative memory. *Psychol Aging* **18**, 149-160 (2003).
- 17 Craik, F. I. M. & Jennings, J. M. in *The Handbook of Aging and Cognition* (eds F.I.M. Craik & T. A. Salthouse) (Lawrence Erlbaum Associates, 1992).
- 18 Bäckman, L., Small, B. J. & Wahlin, A. in *Handbook of aging and cognition II* Vol. 499-558 (eds F. I. M. Craik & T. A. Salthouse) (Erlbaum, 2000).
- 19 Balota, D. A., Dolan, P. O. & Duchek, J. M. in *Oxford Handbook of Memory* (eds E. Tulving & F. I. M. Craik) 410 (Oxford University Press, 2000).
- 20 Spaniol, J., Madden, D. J. & Voss, A. A diffusion model analysis of adult age differences in episodic and semantic long-term memory retrieval. *J. Exp. Psychol.-Learn. Mem. Cogn.* **32**, 101-117, doi:10.1037/0278-7393.1.101 (2006).
- 21 Pudas, S., Persson, J., Nilsson, L. G. & Nyberg, L. Midlife memory ability accounts for brain activity differences in healthy aging. *Neurobiol Aging* **35**, 2495-2503, doi:10.1016/j.neurobiolaging.2014.05.022 (2014).
- 22 Pudas, S., Persson, J., Josefsson, M., de Luna, X., Nilsson, L. G. & Nyberg, L. Brain characteristics of individuals resisting age-related cognitive decline over two decades. *The Journal of neuroscience : the official journal of the Society for Neuroscience* **33**, 8668-8677, doi:10.1523/JNEUROSCI.2900-12.2013

(2013).

- 23 Zaretsky, H. H. & Halberstam, J. L. Effects of aging, brain damage, and associative strength on paired-associate learning and relearning. *Journal of Genetic Psychology* **112**, 165-168 (1968).
- 24 Kausler, D. H. & Lair, C. V. Associative strength and paired-associate learning in elderly subjects. *Journal of Gerontology* **21**, 278-280 (1966).
- 25 Zaretsky, H. H. & Halberstam, J. L. Age differences in paired-associate learning. *Journal of Gerontology* **23**, 165-168 (1968).
- 26 Gardini, S., Cuetos, F., Fasano, F., Pellegrini, F. F., Marchi, M., Venneri, A. & Caffarra, P. Brain structural substrates of semantic memory decline in mild cognitive impairment. *Curr Alzheimer Res* **10**, 373-389 (2013).
- 27 Gardini, S., Venneri, A., Sambataro, F., Cuetos, F., Fasano, F., Marchi, M., Crisi, G. & Caffarra, P. Increased functional connectivity in the default mode network in mild cognitive impairment: a maladaptive compensatory mechanism associated with poor semantic memory performance. *J Alzheimers Dis* **45**, 457-470, doi:10.3233/JAD-142547 (2015).
- 28 Hirni, D. I., Kivisaari, S. L., Monsch, A. U. & Taylor, K. I. Distinct neuroanatomical bases of episodic and semantic memory performance in Alzheimer's disease. *Neuropsychologia* **51**, 930-937, doi:10.1016/j.neuropsychologia.2013.01.013 (2013).
- 29 Killiany, R. J., Hyman, B. T., Gomez-Isla, T., Moss, M. B., Kikinis, R., Jolesz, F., Tanzi, R., Jones, K. & Albert, M. S. MRI measures of entorhinal cortex vs hippocampus in preclinical AD. *Neurology* **58**, 1188-1196 (2002).
- 30 Desikan, R. S., Cabral, H. J., Hess, C. P., Dillon, W. P., Glastonbury, C. M., Weiner, M. W., Schmansky, N. J., Greve, D. N., Salat, D. H., Buckner, R. L., Fischl, B. & Alzheimers Dis, N. Automated MRI measures identify individuals with mild cognitive impairment and Alzheimers disease. *Brain* **132**, 2048-2057, doi:10.1093/brain/awp123 (2009).
- 31 Augustinack, J. C., Huber, K. E., Stevens, A. A., Roy, M., Frosch, M. P., van der Kouwe, A. J. W., Wald, L. L., Van Leemput, K., McKee, A. C., Fischl, B. & Alzheimers Dis Neuroimaging, I. Predicting the location of human perirhinal cortex, Brodmann's area 35, from MRI. *Neuroimage* **64**, 32-42, doi:10.1016/j.neuroimage.2012.08.071 (2013).
- 32 Jauhiainen, A. M., Pihlajamäki, M., Tervo, S., Niskanen, E., Tanila, H., Hanninen, T., Vanninen, R. L. & Soininen, H. Discriminating Accuracy of Medial Temporal Lobe Volumetry and fMRI in Mild Cognitive Impairment. *Hippocampus* **19**, 166-175, doi:10.1002/hipo.20494 (2009).
- 33 Duara, R., Loewenstein, D. A., Potter, E., Appel, J., Greig, M. T., Urs, R., Shen, Q., Raj, A., Small, B., Barker, W., Schofield, E., Wu, Y. & Potter, H. Medial temporal lobe atrophy on MRI scans and the diagnosis of Alzheimer disease. *Neurology* **71**, 1986-1992, doi:10.1212/01.wnl.0000336925.79704.9f (2008).
- 34 Kausler, D. H. *Learning and memory in normal aging.*, (Academic Press, 1994).
- 35 Healy, G., Gurin, C. & Smeaton, A. F. in *Quantified Self Europe Conference* (Amsterdam, Netherlands, 2014).
- 36 Naveh-Benjamin, M. Adult age differences in memory performance: Tests of an associative deficit hypothesis. *Journal of Experimental Psychology: Learning, Memory, & Cognition* **26**, 1170-1187 (2000).
- 37 Froger, C., Tacconat, L., Landre, L., Beigneux, K. & Isingrini, M. Effects of level of processing at encoding and types of retrieval task in mild cognitive impairment and normal aging. *J. Clin. Exp. Neuropsychol.* **31**, 312-321, doi:10.1080/13803390802112554 (2009).
- 38 Hudon, C., Villeneuve, S. & Belleville, S. The effect of semantic orientation at encoding on free-recall performance in amnesic mild cognitive impairment and probable Alzheimer's disease. *J. Clin. Exp. Neuropsychol.* **33**, 631-638, doi:10.1080/13803395.2010.547663 (2011).
- 39 Mandzia, J. L., McAndrews, M. P., Grady, C. L., Graham, S. J. & Black, S. E. Neural correlates of incidental memory in mild cognitive impairment: An fMRI study. *Neurobiol. Aging* **30**, 717-730, doi:10.1016/j.neurobiolaging.2007.08.024 (2009).
- 40 Petersen, R. C., Smith, G. E., Waring, S. C., Ivnik, R. J., Tangalos, E. G. & Kokmen, E. Mild cognitive impairment: clinical characterization and outcome. *Arch Neurol* **56**, 303-308 (1999).
- 41 Owsley, C. Aging and vision. *Vision research* **51**, 1610-1622 (2011).
- 42 Sekuler, R. & Sekuler, A. B. in *Encyclopedia of Psychology* Vol. 8 (ed A Kazdin) 180-183 (American Psychological Association and Cambridge University Press, 2000).
- 43 Peters, A., Moss, M. B. & Sethares, C. The effects of aging on layer 1 of primary visual cortex in the rhesus monkey. *Cerebral cortex* **11**, 93-103 (2001).
- 44 Peters, A., Sethares, C. & Killiany, R. J. Effects of age on the thickness of myelin sheaths in monkey primary visual cortex. *J. Comp. Neurol.* **435**, 241-248 (2001).
- 45 Raz, N., Lindenberger, U., Rodrigue, K. M., Kennedy, K. M., Head, D., Williamson, A., Dahle, C., Gerstorf,

- D. & Acker, J. D. Regional Brain Changes in Aging Healthy Adults: General Trends, Individual Differences and Modifiers. *Cerebral cortex* (2005).
- 46 Bublak, P., Redel, P., Sorg, C., Kurz, A., Forstl, H., Muller, H. J., Schneider, W. X. & Finke, K. Staged decline of visual processing capacity in mild cognitive impairment and Alzheimer's disease. *Neurobiol. Aging* **32**, 1219-1230, doi:10.1016/j.neurobiolaging.2009.07.012 (2011).
- 47 Javaid, F. Z., Brenton, J., Guo, L. & Cordeiro, M. F. Visual and Ocular Manifestations of Alzheimer's Disease and Their Use as Biomarkers for Diagnosis and Progression. *Front. Neurol.* **7**, 11, doi:10.3389/fneur.2016.00055 (2016).
- 48 Cogan, D. G. VISUAL DISTURBANCES WITH FOCAL PROGRESSIVE DEMENTING DISEASE. *Am. J. Ophthalmol.* **100**, 68-72, doi:10.1016/s0002-9394(14)74985-2 (1985).
- 49 Uhlmann, R. F., Larson, E. B., Koepsell, T. D., Rees, T. S. & Duckert, L. G. VISUAL IMPAIRMENT AND COGNITIVE DYSFUNCTION IN ALZHEIMERS-DISEASE. *J. Gen. Intern. Med.* **6**, 126-132, doi:10.1007/bf02598307 (1991).
- 50 Croningolomb, A., Corkin, S., Rizzo, J. F., Cohen, J., Growdon, J. H. & Banks, K. S. VISUAL DYSFUNCTION IN ALZHEIMERS-DISEASE - RELATION TO NORMAL AGING. *Annals of Neurology* **29**, 41-52, doi:10.1002/ana.410290110 (1991).
- 51 Pache, M., Smeets, C. H. W., Gasio, P. F., Savaskan, E., Flammer, J., Wirz-Justice, A. & Kaiser, H. J. Colour vision deficiencies in Alzheimer's disease. *Age Ageing* **32**, 422-426, doi:10.1093/ageing/32.4.422 (2003).
- 52 Gilmore, G. C., Wenk, H. E., Naylor, L. A. & Koss, E. MOTION PERCEPTION AND ALZHEIMERS-DISEASE. *J. Gerontol.* **49**, P52-P57, doi:10.1093/geronj/49.2.P52 (1994).
- 53 Mielke, R., Kessler, J., Fink, G., Herholz, K. & Heiss, W. D. DYSFUNCTION OF VISUAL-CORTEX CONTRIBUTES TO DISTURBED PROCESSING OF VISUAL INFORMATION IN ALZHEIMERS-DISEASE. *Int. J. Neurosci.* **82**, 1-9, doi:10.3109/00207459508994285 (1995).
- 54 Cronin-Golomb, A., Corkin, S. & Growdon, J. H. Visual Dysfunction Predicts Cognitive Deficits in Alzheimers-Disease. *Optom. Vis. Sci.* **72**, 168-176 (1995).
- 55 Bouras, C., Hof, P. R., Giannakopoulos, P., Michel, J. P. & Morrison, J. H. Regional Distribution of Neurofibrillary Tangles and Senile Plaques in the Cerebral-Cortex of Elderly Patients - a Quantitative-Evaluation of a One-Year Autopsy Population from a Geriatric Hospital. *Cerebral cortex* **4**, 138-150 (1994).
- 56 Arnold, S. E., Hyman, B. T., Flory, J., Damasio, A. R. & Van Hoesen, G. W. The Topographical and Neuroanatomical Distribution of Neurofibrillary Tangles and Neuritic Plaques in the Cerebral Cortex of Patients with Alzheimer's Disease. *Cerebral cortex* **1**, 103-116 (1991).
- 57 McKee, A. C., Au, R., Cabral, H. J., Kowall, N. W., Seshadri, S., Kubilus, C. A. & Wolf, P. A. Visual association pathology in preclinical Alzheimer disease. *J. Neuropathol. Exp. Neurol.* **65**, 621-630 (2006).
- 58 Belleville, S., Fouquet, C., Hudon, C., Zomahoun, H. T. V., Croteau, J. & Consortium Early, I. Neuropsychological Measures that Predict Progression from Mild Cognitive Impairment to Alzheimer's type dementia in Older Adults: a Systematic Review and Meta-Analysis. *Neuropsychol. Rev.* **27**, 328-353, doi:10.1007/s11065-017-9361-5 (2017).
- 59 Didic, M., Barbeau, E. J., Felician, O., Tramon, E., Guedj, E., Poncet, M. & Ceccaldi, M. Which Memory System is Impaired First in Alzheimer's Disease? *J. Alzheimers Dis.* **27**, 11-22, doi:10.3233/jad-2011-110557 (2011).
- 60 Kirchberg, B. C., Cohen, J. R., Adelsky, M. B., Buthorn, J. J., Gomar, J. J., Gordon, M., Koppel, J., Christen, E., Conejero-Goldberg, C., Davies, P. & Goldberg, T. E. Semantic Distance Abnormalities in Mild Cognitive Impairment: Their Nature and Relationship to Function. *Am. J. Psychiat.* **169**, 1275-1283, doi:10.1176/appi.ajp.2012.12030383 (2012).
- 61 Kim, J. E., Park, S. H., Hong, Y. J., Hwang, J., Han, N. E., Lee, S. M., Roh, J. H., Kim, J. S. & Lee, J. H. Qualitative Comparison of Semantic Memory Impairment in Patients with Amnesic Mild Cognitive Impairment Based on beta-Amyloid Status. *J. Clin. Neurol.* **15**, 27-37, doi:10.3988/jcn.2019.15.1.27 (2019).
- 62 Venneri, A., Jahn-Carta, C., De Marco, M., Quaranta, D. & Marra, C. Diagnostic and prognostic role of semantic processing in preclinical Alzheimer's disease. *Biomark. Med.* **12**, 637-651, doi:10.2217/bmm-2017-0324 (2018).
- 63 Joubert, S., Brambati, S. M., Ansado, J., Barbeau, E. J., Felician, O., Didic, M., Lacombe, J., Goldstein, R., Chayer, C. & Kergoat, M. J. The cognitive and neural expression of semantic memory impairment in mild cognitive impairment and early Alzheimer's disease. *Neuropsychologia* **48**, 978-988, doi:10.1016/j.neuropsychologia.2009.11.019 (2010).
- 64 Kirchhoff, B. A., Anderson, B. A., Barch, D. M. & Jacoby, L. L. Cognitive and Neural Effects of Semantic Encoding Strategy Training in Older Adults. *Cerebral cortex* **22**, 788-799, doi:10.1093/cercor/bhr129 (2012).

- 65 Belleville, S., Gilbert, B., Fontaine, F., Gagnon, L., Menard, E. & Gauthier, S. Improvement of episodic memory in persons with mild cognitive impairment and healthy older adults: Evidence from a cognitive intervention program. *Dement. Geriatr. Cogn. Disord.* **22**, 486-499, doi:10.1159/000096316 (2006).
- 66 Park, D. C., Polk, T., Mikels, J. A., Taylor, S. F. & Marshuetz, C. Cerebral aging: Integration of brain and behavioral models of cognitive function. *Dialogues in Clinical Neuroscience* **3**, 151-164 (2001).
- 67 Verhaeghen, P. Aging and vocabulary scores: A meta-analysis. *Psychol. Aging* **18**, 332-339, doi:10.1037/0882-7974.18.2.332 (2003).
- 68 Spreng, R. N. & Turner, G. R. The shifting architecture of cognition and brain function in older adulthood. *Perspectives on Psychological Science* (2019).
- 69 Umanath, S. & Marsh, E. J. Understanding how prior knowledge influences memory in older adults. *Perspectives on Psychological Science* **9**, 408-426 (2014).
- 70 Badre, D. & Wagner, A. D. Left ventrolateral prefrontal cortex and the cognitive control of memory. *Neuropsychologia* **45**, 2883-2901 (2007).
- 71 Ranganath, C. & Ritchey, M. Two cortical systems for memory-guided behaviour. *Nature Reviews Neuroscience* **13**, 713-726, doi:10.1038/nrn3338 (2012).
- 72 Moscovitch, M., Cabeza, R., Winocur, G. & Nadel, L. Episodic Memory and Beyond: The Hippocampus and Neocortex in Transformation. *Annual Review of Psychology*, Vol 67 **67**, 105-+, doi:10.1146/annurev-psych-113011-143733 (2016).
- 73 Kahn, I., Andrews-Hanna, J. R., Vincent, J. L., Snyder, A. Z. & Buckner, R. L. Distinct cortical anatomy linked to subregions of the medial temporal lobe revealed by intrinsic functional connectivity. *Journal of Neurophysiology* **100**, 129-139, doi:10.1152/jn.00077.2008 (2008).
- 74 Libby, L. A., Ekstrom, A. D., Ragland, J. D. & Ranganath, C. Differential connectivity of perirhinal and parahippocampal cortices within human hippocampal subregions revealed by high-resolution functional imaging. *The Journal of neuroscience : the official journal of the Society for Neuroscience* **32**, 6550-6560, doi:10.1523/JNEUROSCI.3711-11.2012 (2012).
- 75 Roediger, H. L., Weldon, M. S. & Challis, B. H. in *Varieties of memory and consciousness: Essays in honour of Endel Tulving*. (eds H. L. Roediger & F. I. M. Craik) 3-41 (Erlbaum, 1989).
- 76 Roediger, H. L. & McDermott, K. B. in *Handbook of Neuropsychology* Vol. 8 (eds F. Boller & J. Grafman) (Elsevier, 1993).
- 77 Tulving, E. in *The cognitive neurosciences* (ed M. S. Gazzaniga) (MIT Press, 1995).
- 78 Bahrack, H. P. & Boucher, B. Retention of Visual and Verbal Codes of Same Stimuli. *Journal of experimental psychology* **78**, 417-& (1968).
- 79 Bahrack, H. P. & Bahrack, P. Independence of verbal and visual codes of the same stimuli. *Journal of experimental psychology* **91**, 344-346 (1971).
- 80 Blumenthal, A., Stojanoski, B., Martin, C. B., Cusack, R. & Kohler, S. Animacy and real-world size shape object representations in the human medial temporal lobes. *Hum. Brain Mapp.* **39**, 3779-3792, doi:10.1002/hbm.24212 (2018).
- 81 Clarke, A., Taylor, K. I., Devereux, B., Randall, B. & Tyler, L. K. From perception to conception: how meaningful objects are processed over time. *Cerebral cortex* **23**, 187-197, doi:10.1093/cercor/bhs002 (2013).
- 82 Clarke, A. & Tyler, L. K. Object-Specific Semantic Coding in Human Perirhinal Cortex. *J. Neurosci.* **34**, 4766-4775, doi:10.1523/jneurosci.2828-13.2014 (2014).
- 83 Clarke, A. & Tyler, L. K. Understanding What We See: How We Derive Meaning From Vision. *Trends Cogn. Sci.* **19**, 677-687, doi:10.1016/j.tics.2015.08.008 (2015).
- 84 Devereux, B. J., Clarke, A., Marouchos, A. & Tyler, L. K. Representational Similarity Analysis Reveals Commonalities and Differences in the Semantic Processing of Words and Objects. *J. Neurosci.* **33**, 18906-18916, doi:10.1523/jneurosci.3809-13.2013 (2013).
- 85 Davis, S. W., Wang, W. C., Wing, E. A., Geib, B. R. & Cabeza, R. Contribution of visual and semantic objects representations to subsequent perceptual and conceptual memory. (in preparation).
- 86 Alvarez, P. & Squire, L. R. Memory consolidation and the medial temporal lobe: a simple network model. *Proc. Natl. Acad. Sci. U. S. A.* **91**, 7041-7045 (1994).
- 87 Norman, K. A. & O'Reilly, R. C. Modeling hippocampal and neocortical contributions to recognition memory: a complementary-learning-systems approach. *Psychol Rev* **110**, 611-646 (2003).
- 88 Rugg, M. D., Johnson, J. D., Park, H. & Uncapher, M. R. in *Prog. Brain Res.* Vol. 169 *Progress in Brain Research* (eds W. S. Sossin, J. C. Lacaille, V. F. Castellucci, & S. Belleville) 339-352 (Elsevier Science Bv, 2008).
- 89 Danker, J. F. & Anderson, J. R. The Ghosts of Brain States Past: Remembering Reactivates the Brain Regions Engaged During Encoding. *Psychol. Bull.* **136**, 87-102, doi:10.1037/a0017937 (2010).
- 90 Wing, E. A., Ritchey, M. & Cabeza, R. Reinstatement of Individual Past Events Revealed by the Similarity

of Distributed Activation Patterns during Encoding and Retrieval. *J. Cogn. Neurosci.* **27**, 679-691, doi:10.1162/jocn_a_00740 (2015).

- 91 Payer, D., Marshuetz, C., Sutton, B., Hebrank, A., Welsh, R. C. & Park, D. C. Decreased neural specialization in old adults on a working memory task. *Neuroreport* **17**, 487-491, doi:10.1097/01.wnr.0000209005.40481.31 (2006).
- 92 Dennis, N. A., Hayes, S. M., Prince, S. E., Madden, D. J., Huettel, S. A. & Cabeza, R. Effects of aging on the neural correlates of successful item and source memory encoding. *J Exp Psychol Learn Mem Cogn* **34**, 791-808 (2008).
- 93 Kung, C. C., Peissig, J. J. & Tarr, M. J. Is region-of-interest overlap comparison a reliable measure of category specificity? *J. Cogn. Neurosci.* **19**, 2019-2034 (2007).
- 94 Wang, T. H., Johnson, J. D., de Chastelaine, M., Donley, B. E. & Rugg, M. D. The Effects of Age on the Neural Correlates of Recollection Success, Recollection-Related Cortical Reinstatement, and Post-Retrieval Monitoring. *Cerebral cortex* **26**, 1698-1714, doi:10.1093/cercor/bhu333 (2016).
- 95 St-Laurent, M., Abdi, H., Bondad, A. & Buchsbaum, B. R. Memory Reactivation in Healthy Aging: Evidence of Stimulus-Specific Dedifferentiation. *J. Neurosci.* **34**, 4175-4186, doi:10.1523/jneurosci.3054-13.2014 (2014).
- 96 Bowman, C. R., Chamberlain, J. D. & Dennis, N. A. Sensory Representations Supporting Memory Specificity: Age Effects on Behavioral and Neural Discriminability. *J. Neurosci.* **39**, 2265-2275, doi:10.1523/jneurosci.2022-18.2019 (2019).
- 97 Monge, Z. A., Wing, E. A., Geib, B. R., Davis, S. W. & Cabeza, R. Age-related dedifferentiation and hyperdifferentiation of perceptual and mnemonic representations. (under review).
- 98 Kriegeskorte, N. Deep Neural Networks: A New Framework for Modeling Biological Vision and Brain Information Processing. *Annual Review of Vision Science, Vol 1* **1**, 417-446, doi:10.1146/annurev-vision-082114-035447 (2015).
- 99 Gauthier, I. & Tarr, M. J. in *Annual Review of Vision Science, Vol 2* Vol. 2 *Annual Review of Vision Science* (eds J. A. Movshon & B. A. Wandell) 377-396 (Annual Reviews, 2016).
- 100 Wen, H., Shi, J., Zhang, Y., Lu, K. H., Cao, J. & Liu, Z. Neural encoding and decoding with deep Learning for dynamic natural vision. *Cerebral cortex*, 1-25, doi:10.1093/cercor/bhx268 (2017).
- 101 Kriegeskorte, N. Deep neural networks: a new framework for modeling biological vision and brain information processing. *Annu Rev Vis Sci* **1**, 417-446, doi:10.1146/annurev-vision-082114-035447 (2015).
- 102 Daselaar, S. M., Fleck, M., Dobbins, I. G., Madden, D. J. & Cabeza, R. Effects of healthy aging on hippocampal and rhinal memory functions: An event-related fMRI study. *Cerebral cortex* **16**, 1771-1782 (2006).
- 103 Giovanello, K. S., De Brigard, F., Hennessey Ford, J., Kaufer, D. I., Burke, J. R., Browndyke, J. N. & Welsh-Bohmer, K. A. Event-related functional magnetic resonance imaging changes during relational retrieval in normal aging and amnesic mild cognitive impairment. *Journal of the International Neuropsychological Society : JINS* **18**, 886-897, doi:10.1017/S1355617712000689 (2012).
- 104 Juottonen, K., Laakso, M. P., Insausti, R., Lehtovirta, M., Pitkanen, A., Partanen, K. & Soininen, H. Volumes of the entorhinal and perirhinal cortices in Alzheimer's disease. *Neurobiol. Aging* **19**, 15-22, doi:10.1016/s0197-4580(98)00007-4 (1998).
- 105 Teipel, S. J., Pruessner, J. C., Faltraco, F., Born, C., Rocha-Unold, M., Evans, A., Moller, H. J. & Hampel, H. Comprehensive dissection of the medial temporal lobe in AD: measurement of hippocampus, amygdala, entorhinal, perirhinal and parahippocampal cortices using MRI. *Journal of Neurology* **253**, 794-800, doi:10.1007/s00415-006-0120-4 (2006).
- 106 Braak, H. & Braak, E. On areas of transition between entorhinal allocortex and temporal isocortex in the human brain. Normal morphology and lamina-specific pathology in Alzheimer's disease. *Acta Neuropathol* **68**, 325-332 (1985).
- 107 Taylor, K. I. & Probst, A. Anatomic localization of the transentorhinal region of the perirhinal cortex. *Neurobiol. Aging* **29**, 1591-1596, doi:10.1016/j.neurobiolaging.2007.03.024 (2008).
- 108 Braak, H. & Braak, E. Frequency of stages of Alzheimer-related lesions in different age categories. *Neurobiol Aging* **18**, 351-357 (1997).
- 109 Carstensen, L. L., Fung, H. H. & Charles, S. T. Socioemotional selectivity theory and the regulation of emotion in the second half of life. *Motivation and Emotion* **27**, 103-123 (2003).
- 110 Li, Y., Baldassi, M., Johnson, E. J. & Weber, E. U. Complementary Cognitive Capabilities, Economic Decision Making, and Aging. *Psychol. Aging* **28**, 595-613, doi:10.1037/a0034172 (2013).
- 111 Devitt, A. L. & Schacter, D. L. False memories with age: Neural and cognitive underpinnings. *Neuropsychologia* **91**, 346-359 (2016).
- 112 Arbuckle, T. Y., Cooney, R., Milne, J. & Melchior, A. Memory for spatial layouts in relation to age and schema typicality. *Psychol. Aging* **9**, 467 (1994).

- 113 Quaranta, D., Vita, M. G., Spinelli, P., Scaricamazza, E., Castelli, D., Lacidogna, G., Piccininni, C., Rossini, P. M., Gainotti, G. & Marra, C. Does Semantic Memory Impairment in Amnesic aMCI with Hippocampal Atrophy Conform to a Distinctive Pattern of Progression? *Current Alzheimer Research* **11**, 399-407 (2014).
- 114 Barulli, D. & Stern, Y. Efficiency, capacity, compensation, maintenance, plasticity: emerging concepts in cognitive reserve. *Trends Cogn. Sci.* **17**, 502-509, doi:10.1016/j.tics.2013.08.012 (2013).
- 115 Craik, F. I. M. & Rose, N. S. Memory encoding and aging: A neurocognitive perspective. *Neurosci. Biobehav. Rev.* **36**, 1729-1739, doi:10.1016/j.neubiorev.2011.11.007 (2012).
- 116 Otten, L. J., Henson, R. N. & Rugg, M. D. Depth of processing effects on neural correlates of memory encoding: Relationship between findings from across- and within-task comparisons. *Brain* **124**, 399-412 (2001).
- 117 Preston, A. R. & Eichenbaum, H. Interplay of Hippocampus and Prefrontal Cortex in Memory. *Curr. Biol.* **23**, R764-R773, doi:10.1016/j.cub.2013.05.041 (2013).
- 118 van Kesteren, M. T., Ruiter, D. J., Fernandez, G. & Henson, R. N. How schema and novelty augment memory formation. *Trends Neurosci* **35**, 211-219, doi:10.1016/j.tins.2012.02.001 (2012).
- 119 Wang, S. H. & Morris, R. G. M. in *Annual Review of Psychology* Vol. 61 *Annual Review of Psychology* 49-79 (Annual Reviews, 2010).
- 120 Schneider, B. A. & Pichora-Fuller, M. K. in *Handbook of Cognitive Aging II* (eds F. I. M. Craik & T. A. Salthouse) 155-219 (Erlbaum, 2000).
- 121 Monge, Z. A. & Madden, D. J. Linking cognitive and visual perceptual decline in healthy aging: The information degradation hypothesis. *Neurosci. Biobehav. Rev.* **69**, 166-173, doi:10.1016/j.neubiorev.2016.07.031 (2016).
- 122 Gilmore, G. C., Spinks, R. A. & Thomas, C. W. Age effects in coding tasks: Componential analysis and test of the sensory deficit hypothesis. *Psychol. Aging* **21**, 7-18, doi:10.1037/0882-7974.21.1.7 (2006).
- 123 Murphy, D. R., Craik, F. I., Li, K. Z. & Schneider, B. A. Comparing the effects of aging and background noise on short-term memory performance. *Psychol Aging* **15**, 323-334 (2000).
- 124 Pichora-Fuller, M. K., Schneider, B. A. & Daneman, M. How Young and Old Adults Listen to and Remember Speech in Noise. *Journal of the Acoustical Society of America* **97**, 593-608 (1995).
- 125 Craik, F. I. M. & Lockhart, R. S. Levels of processing: A framework for memory research. *Journal of Verbal Learning and Verbal Behavior* **11**, 671-684 (1972).
- 126 Craik, F. I. M. & Tulving, E. Depth of processing and the retention of words in episodic memory. *Journal of Experimental Psychology: General* **104**, 268-294 (1975).
- 127 Troyer, A. K., Hafliger, A., Cadieux, M. J. & Craik, F. I. M. Name and face learning in older adults: effects of level of processing, self-generation, and intention to learn. *J. Gerontol. Ser. B-Psychol. Sci. Soc. Sci.* **61**, P67-P74, doi:10.1093/geronb/61.2.P67 (2006).
- 128 Ritchey, M., Bessette-Symons, B., Hayes, S. M. & Cabeza, R. Emotion processing in the aging brain is modulated by semantic elaboration. *Neuropsychologia* **49**, 640-650, doi:S0028-3932(10)00389-1 [pii] 10.1016/j.neuropsychologia.2010.09.009 (2011).
- 129 Mandzia, J. L., Black, S. E., McAndrews, M. P., Grady, C. & Graham, S. fMRI differences in encoding and retrieval of pictures due to encoding strategy in the elderly. *Hum. Brain Mapp.* **21**, 1-14, doi:10.1002/hbm.10140 (2004).
- 130 Konkle, T., Brady, T. F., Alvarez, G. A. & Oliva, A. Conceptual Distinctiveness Supports Detailed Visual Long-Term Memory for Real-World Objects. *J. Exp. Psychol.-Gen.* **139**, 558-578, doi:10.1037/a0019165 (2010).
- 131 Old, S. R. & Naveh-Benjamin, M. Differential effects of age on item and associative measures of memory: A meta-analysis. *Psychol. Aging* **23**, 104-118, doi:Doi 10.1037/0882-7974.23.1.104 (2008).
- 132 Wang, W. C. & Cabeza, R. in *Cognitive Neuroscience of Aging - Second Edition* (eds R. Cabeza, L. Nyberg, & D. Park) (Oxford University Press, 2016).
- 133 Pini, L., Pievani, M., Bocchetta, M., Altomare, D., Bosco, P., Cavedo, E., Galluzzi, S., Marizzoni, M. & Frisoni, G. B. Brain atrophy in Alzheimer's Disease and aging. *Ageing Res. Rev.* **30**, 25-48, doi:10.1016/j.arr.2016.01.002 (2016).
- 134 Pan, P. L., Zhu, L., Yu, T. T., Shi, H. C., Zhang, B., Qin, R. M., Zhu, X. L., Qian, L., Zhao, H., Zhou, H. & Xu, Y. Aberrant spontaneous low-frequency brain activity in amnesic mild cognitive impairment: A meta-analysis of resting-state fMRI studies. *Ageing Res. Rev.* **35**, 12-21, doi:10.1016/j.arr.2016.12.001 (2017).
- 135 Naveh-Benjamin, M., Hussain, Z., Guez, J. & Bar-On, M. Adult age differences in episodic memory: further support for an associative-deficit hypothesis. *J Exp Psychol Learn Mem Cogn* **29**, 826-837 (2003).
- 136 Naveh-Benjamin, M., Craik, F. I., Guez, J. & Kreuger, S. Divided attention in younger and older adults: effects of strategy and relatedness on memory performance and secondary task costs. *J Exp Psychol*

Learn Mem Cogn **31**, 520-537 (2005).

- 137 Badham, S. P., Estes, Z. & Maylor, E. A. Integrative and Semantic Relations Equally Alleviate Age-Related
Associative Memory Deficits. *Psychol. Aging* **27**, 141-152, doi:10.1037/a0023924 (2012).
- 138 Badham, S. P. & Maylor, E. A. Antimnemonic effects of schemas in young and older adults. *Aging
Neuropsychol. Cogn.* **23**, 78-102, doi:10.1080/13825585.2015.1048774 (2016).
- 139 Castel, A. D. Memory for grocery prices in younger and older adults: The role of schematic support.
Psychol. Aging **20**, 718-721, doi:10.1037/0882-7974.20.4.718 (2005).
- 140 Shi, L. Z., Tang, W. H. & Liu, X. P. Age-related schema reliance of judgments of learning in predicting
source memory. *Aging Neuropsychol. Cogn.* **19**, 301-318, doi:10.1080/13825585.2011.632616 (2012).
- 141 Atienza, M., Atalaia-Silva, K. C., Gonzalez-Escamilla, G., Gil-Neciga, E., Suarez-Gonzalez, A. & Cantero,
J. L. Associative memory deficits in mild cognitive impairment: The role of hippocampal formation.
Neuroimage **57**, 1331-1342, doi:10.1016/j.neuroimage.2011.05.047 (2011).
- 142 Perri, R., Carlesimo, G. A., Serra, L., Caltagirone, C. & Italian Interdisciplinary, N. When the Amnesic
Mild Cognitive Impairment Disappears Characterisation of the Memory Profile. *Cogn. Behav. Neurol.*
22, 109-116, doi:10.1097/WNN.ob013e3181a7225c (2009).
- 143 Frankland PW & Bontempi B. Fast track to the medial prefrontal cortex. *Proc Natl Acad Sci USA* **103**,
509-510 (2006).
- 144 van Kesteren, M. T. R., Rijpkema, M., Ruiter, D. J. & Fernandez, G. Retrieval of Associative Information
Congruent with Prior Knowledge Is Related to Increased Medial Prefrontal Activity and Connectivity. *J.*
Neurosci. **30**, 15888-15894 (2010).
- 145 Wing, E. A., Iyengar, V., Van Kesteren, M. T. R. & Cabeza, R. Schema-congruent memory benefit is
associated with mPFC-mediated reactivation of neocortical memory representations. (under review).
- 146 Dennis, N. A. & Cabeza, R. in *Handbook of Aging and Cognition - III* (eds F. I. M. Craik & T. A.
Salthouse) (Erlbaum, 2008).
- 147 Fjell, A. M., Westlye, L. T., Grydeland, H., Amlien, I., Espeseth, T., Reinvang, I., Raz, N., Dale, A. M.,
Walhovd, K. B. & Alzheimer Dis Neuroimaging, I. Accelerating Cortical Thinning: Unique to Dementia
or Universal in Aging? *Cerebral cortex* **24**, 919-934, doi:10.1093/cercor/bhs379 (2014).
- 148 Fjell, A. M., Westlye, L. T., Amlien, I., Espeseth, T., Reinvang, I., Raz, N., Agartz, I., Salat, D. H., Greve,
D. N., Fischl, B., Dale, A. M. & Walhovd, K. B. High Consistency of Regional Cortical Thinning in Aging
across Multiple Samples. *Cerebral cortex* **19**, 2001-2012 (2009).
- 149 Sowell, E. R., Peterson, B. S., Thompson, P. M., Welcome, S. E., Henkenius, A. L. & Toga, A. W. Mapping
cortical change across the human life span. *Nat. Neurosci.* **6**, 309-315, doi:10.1038/nn1008 (2003).
- 150 Lighthall, N. R., Huettel, S. A. & Cabeza, R. Functional compensation in the ventromedial prefrontal
cortex improves memory-dependent decisions in older adults. *The Journal of neuroscience : the official
journal of the Society for Neuroscience* **34**, 15648-15657, doi:10.1523/JNEUROSCI.2888-14.2014
(2014).
- 151 Bar, M. & Bubic, A. in *Oxford Handbook of Cognitive Neurosciences, vol. 1* (ed Kevin Ochsner and
Stephen Kosslyn) (Oxford University Press, 2013).
- 152 Gazzaley, A. & Nobre, A. C. Top-down modulation: bridging selective attention and working memory.
Trends Cogn. Sci. **16**, 129-135, doi:10.1016/j.tics.2011.11.014 (2012).
- 153 O'Craven, K. M. & Kanwisher, N. Mental imagery of faces and places activates corresponding stimulus-
specific brain regions. *J. Cogn. Neurosci.* **12**, 1013-1023 (2000).
- 154 Ishai, A., Ungerleider, L. G. & Haxby, J. V. Distributed neural systems for the generation of visual images.
Neuron **28**, 979-990 (2000).
- 155 Esterman, M. & Yantis, S. Perceptual Expectation Evokes Category-Selective Cortical Activity. *Cerebral
cortex* **20**, 1245-1253, doi:10.1093/cercor/bhp188 (2010).
- 156 Bollinger, J., Rubens, M. T., Zanto, T. P. & Gazzaley, A. Expectation-driven changes in cortical functional
connectivity influence working memory and long-term memory performance. *The Journal of
neuroscience : the official journal of the Society for Neuroscience* **30**, 14399-14410, doi:30/43/14399
[pii]
10.1523/JNEUROSCI.1547-10.2010 (2010).
- 157 Bollinger, J., Rubens, M. T., Masangkay, E., Kalkstein, J. & Gazzaley, A. An expectation-based memory
deficit in aging. *Neuropsychologia* **49**, 1466-1475, doi:10.1016/j.neuropsychologia.2010.12.021 (2011).
- 158 Gazzaley, A., Cooney, J. W., Rissman, J. & D'Esposito, M. Top-down suppression deficit underlies
working memory impairment in normal aging. *Nat Neurosci* **8**, 1298-1300 (2005).
- 159 Thomas, C., Moya, L., Avidan, G., Humphreys, K., Jung, K. J., Peterson, M. A. & Behrmann, M. Reduction
in white matter connectivity, revealed by diffusion tensor imaging, may account for age-related changes
in face perception. *J. Cogn. Neurosci.* **20**, 268-284 (2008).
- 160 Miller, B. T., Vytlačil, J., Fegen, D., Pradhan, S. & D'Esposito, M. The Prefrontal Cortex Modulates
Category Selectivity in Human Extrastriate Cortex. *J. Cogn. Neurosci.* **23**, 1-10 (2010).

- 161 Cristescu, T. C., Devlin, J. T. & Nobre, A. C. Orienting attention to semantic categories. *Neuroimage* **33**,
1178-1187 (2006).
- 162 Fernandez, G. & Morris, R. G. M. Memory, Novelty and Prior Knowledge. *Trends in Neurosciences* **41**,
654-659, doi:10.1016/j.tins.2018.08.006 (2018).
- 163 Kafkas, A. & Montaldi, D. How do memory systems detect and respond to novelty? *Neuroscience letters*
680, 60-68, doi:10.1016/j.neulet.2018.01.053 (2018).
- 164 Fernandez, R. S., Boccia, M. M. & Pedreira, M. E. The fate of memory: Reconsolidation and the case of
Prediction Error. *Neurosci. Biobehav. Rev.* **68**, 423-441, doi:10.1016/j.neubiorev.2016.06.004 (2016).
- 165 Henson, R. N. & Gagnepain, P. Predictive, Interactive Multiple Memory Systems. *Hippocampus* **20**,
1315-1326 (2010).
- 166 Kriegeskorte, N. & Kievit, R. A. Representational geometry: integrating cognition, computation, and the
brain. *Trends Cogn. Sci.* **17**, 401-412, doi:10.1016/j.tics.2013.06.007 (2013).
- 167 Kriegeskorte, N., Mur, M. & Bandettini, P. Representational similarity analysis—connecting the branches
of systems neuroscience. *Front Syst Neurosci* **2**, 4, doi:10.3389/neuro.06.004.2008 (2008).
- 168 Kriegeskorte, N., Mur, M., Ruff, D. A., Kiani, R., Bodurka, J., Esteky, H., Tanaka, K. & Bandettini, P. A.
Matching Categorical Object Representations in Inferior Temporal Cortex of Man and Monkey. *Neuron*
60, 1126-1141 (2008).
- 169 Petersen, R. C. Clinical practice. Mild cognitive impairment. *N Engl J Med* **364**, 2227-2234,
doi:10.1056/NEJMcip0910237 (2011).
- 170 Kriegeskorte, N., Goebel, R. & Bandettini, P. Information-based functional brain mapping. *Proc Natl
Acad Sci U S A* **103**, 3863-3868, doi:10.1073/pnas.0600244103 (2006).
- 171 Open Science, C. PSYCHOLOGY. Estimating the reproducibility of psychological science. *Science* **349**,
aac4716, doi:10.1126/science.aac4716 (2015).
- 172 Simmons, J. P., Nelson, L. D. & Simonsohn, U. False-positive psychology: undisclosed flexibility in data
collection and analysis allows presenting anything as significant. *Psychol Sci* **22**, 1359-1366,
doi:10.1177/0956797611417632 (2011).
- 173 Poldrack, R. A., Baker, C. I., Durnez, J., Gorgolewski, K. J., Matthews, P. M., Munafò, M. R., Nichols, T.
E., Poline, J. B., Vul, E. & Yarkoni, T. Scanning the horizon: towards transparent and reproducible
neuroimaging research. *Nat Rev Neurosci* **18**, 115-126, doi:10.1038/nrn.2016.167 (2017).
- 174 Button, K. S., Ioannidis, J. P., Mokrysz, C., Nosek, B. A., Flint, J., Robinson, E. S. & Munafò, M. R. Power
failure: why small sample size undermines the reliability of neuroscience. *Nat Rev Neurosci* **14**, 365-376,
doi:10.1038/nrn3475 (2013).
- 175 Erdfeiler, E., Faul, F. & Buchner, A. GPOWER: A general power analysis program. *Behav. Res. Methods
Instr. Comput.* **28**, 1-11, doi:10.3758/bf03203630 (1996).
- 176 Nichols, T. E., Das, S., Eickhoff, S. B., Evans, A. C., Glatard, T., Hanke, M., Kriegeskorte, N., Milham, M.
P., Poldrack, R. A., Poline, J.-B., Proal, E., Thirion, B., Van Essen, D. C., White, T. & Yeo, B. T. T. Best
Practices in Data Analysis and Sharing in Neuroimaging using MRI. *bioRxiv* doi: **10.1101/054262**
(2016).
- 177 Eklund, A., Nichols, T. E. & Knutsson, H. Cluster failure: Why fMRI inferences for spatial extent have
inflated false-positive rates. *Proc. Natl. Acad. Sci. U. S. A.* **113**, 7900-7905, doi:10.1073/pnas.1602413113
(2016).
- 178 Reiss, P. T. Cross-validation and hypothesis testing in neuroimaging: An irenic comment on the exchange
between Friston and Lindquist et al. *Neuroimage* **116**, 248-254, doi:10.1016/j.neuroimage.2015.04.032
(2015).
- 179 Andreano, J. M. & Cahill, L. Sex influences on the neurobiology of learning and memory. *Learn. Mem.*
16, 248-266, doi:10.1101/lm.918309 (2009).
- 180 Li, W. J., van Tol, M. J., Li, M., Miao, W., Jiao, Y. H., Heinze, H. J., Bogerts, B., He, H. G. & Walter, M.
Regional Specificity of Sex Effects on Subcortical Volumes Across the Lifespan in Healthy Aging. *Hum.
Brain Mapp.* **35**, 238-247, doi:10.1002/hbm.22168 (2014).
- 181 Jacobs, E. G., Weiss, B. K., Makris, N., Whitfield-Gabrieli, S., Buka, S. L., Klíbankski, A. & Goldstein, J. M.
Impact of Sex and Menopausal Status on Episodic Memory Circuitry in Early Midlife. *J. Neurosci.* **36**,
10163-10173, doi:10.1523/jneurosci.0951-16.2016 (2016).